Customer logo

Henkel

Functional Design Specification

SampleManager

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chuxiaDocument Signoff				
Name	Position	Company	Signature	Date

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Version Management

Version	Description	Name	Date
1.0	Created	Zhenjuan.Zhang	17 July 2024

2 Introduction

2.1 This Document

2.1.1 Document overview

This document describes the system functionality covered in the Henkel QCLad System Phase A RTM User Requirements Analysis Report.

2.1.2 Disclaimer

The screenshots used below are screenshots generated during the system standard functions or system configuration, as well as customer sample screenshots, the final design may be different from the screenshots in the document, and the data on the screenshots is not the real data of the system, only for reference.

2.1.3 Purpose

This document is used to provide a detailed description of the functionality of the SampleManager system, identifying a description of the system functionality that meets the user's business needs as specified in the User Requirements Analysis Report.

2.1.4 Scope

- Document the functional specifications of the laboratory management system based on the requirements described in the User Requirements Analysis Report.

- Determine the implementation of functionality in the system based on the needs described in the user needs analysis report

2.2 Synopsis

The core application functionality of the QCLad system is the laboratory sample process management functionality, including sample management, SAP interface, task scheduling, master data, reporting, architecture and security.;

2.3 Definitions

The following abbreviations and definitions are assumed throughout this document.

Term/ Abbreviation	Meaning	Description
FS	Function Specification	This Document

Term/ Abbreviation	Meaning	Description
RMB	Right Mouse Button	Using RMB various Sample Manager options can be displayed
QCLAD	SampleManager	Software Name

2.4 References

The following references are made in this document.

Ref. No.	Reference	Description

3 Project Brief

3.1 Project Overview

3.1.1 Project detail

Implementation of SampleManager QCLad will streamline Sample Management, Reports, Dashboards, Instrument and Equipment management, Instrument interfacing and External application integration together with the associated data acquisition across Henkel's labs.

3.2 Functional Requirements

3.2.1 Sample Management

3.2.1.1 Workflow

3.2.1.1.1 Business Scenarios

Generally, in QCLad system, the creation of a sample workflow is based on the creation of a sample lifecycle. According to the process of Henkel, we establish the sample lifecycle workflow, and the user can manually run the workflow and log in the sample as needed, in addition, the workflow can be automatically selected to create the sample by utilizing the SAP interface.

3.2.1.1.2 RTM Mapping

URS Ref.	Requirement	Type
4.1.5.1.1	System shall be able to manage Standard RM test for Incoming workflow. （Standard RM）	Configuration
4.1.5.1.2	System shall be able to manage Standard in process testing for semi finished and finished goods workflow. （without MES)	Configuration
4.1.5.1.3	System shall be able to manage Standard finished goods outgoing test workflow.	Configuration
4.1.5.1.10	System shall be able to manage Test for SCMS/NCMR /CCMS investigation workflow.	Configuration

3.2.1.1.3 Function Design

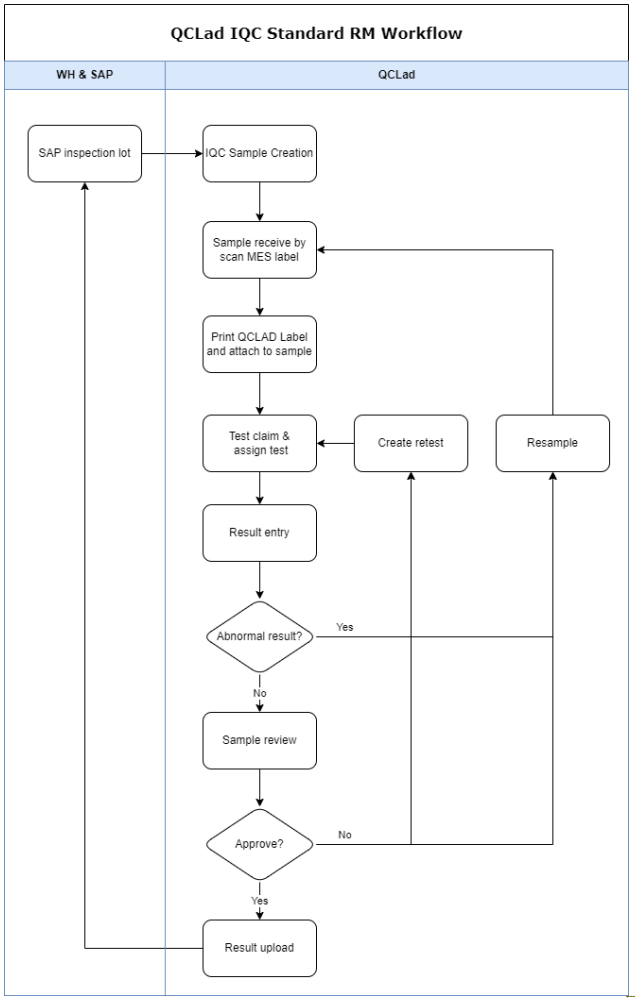
According to user requirements, we have created a QCLad Main Workflow as shown below, based on which different sample workflows can be created for different sample types, QCLad will utilize the standard functionalities and a level of configurations to meet such requirements.

Commented [FS1]: No need supervisor review. Pear review is required no mater within spec or not. Restest should be between result entry and push result back to SAP. Resample should be enabled when sample status are available, in-progress and completed.

Based on user requirements we create four workflows in Phase A as below:

Workflow1: IQC (Standard RM)

According to user requirements, IQC sample workflow is adapt to main workflow, therefore according main workflow system can create IQC(Standard RM) workflow as shown below.



Name	Workflow versi...	Table name	Workflow type	Description	Approval status
IQC (Standard RM)	1	SAMPLE	Sample Login	HENKEL	Authorized

Commented [HH2R1]: Our previous discussion was if the test result is out of spec, we need supervisor review, could you please confirm this is not needed anymore?

Commented [NR3R1]: We have only mentioned on peer review. Again, this is a configuration we are expecting at plant/inspection type level. Currently for Dragon it will be self-approval(test review) and for other plants they may need peer approval. So, we have to maintain the configuration(test review by self in Dragon and for some plants it will be by peer). We may need to change the workflow as there is no need of Supervisor review.

Commented [XW4R1]: Henkel:
review is needed, yet should be optional and configurable on the dimension of "Plant" +"inspection Type (IQC, OQC etc)"

Sample Review:
dragon:
Yuantai:
1)IQC: the same person is allowed to review on the results
2) OQC: the same person is not absolutely not allowed to review on the results.

1)TF will internally evaluate the feasibility as it will touch the core codes.

Regulatory reasons and current process reason.

2) TF to update the workflow diagram to emrge the test review and supervisor review.

Commented [NR5]: In some cases, during resample, the operator can cancel the earlier sample. If the sample is cancelled, the results should be maintained only in QCLAD and should not be pushed to SAP.

Commented [XW6R5]: TF: any sample under Cancel status should not be included in the result interface back to SAP.

Commented [NR7R5]: In PPA, IPQC without MES workflow will not have label from MES. It has to be created within QCLAD.

Workflow2: IPQC (without MES)

According to user requirements, IPQC sample workflow is adapt to main workflow, therefore according main workflow system can create IPQC(without MES) workflow as shown below..

Name	Workflow versi...	Table name	Workflow type	Description	Approval status
IPQC (without MES)	1	SAMPLE	Sample Login	HENKEL	Authorized

Note: Function of IPQC last phase results copy to OQC initial sample results, finial solution is determined to display "copy result" button based on static data IDH that maintains in LIMS. Since it does not affect the IPQC workflow, it will not be described in detail here.

Commented [NR8]: where is the workflow of IPQC? Please make sure the IPQC inspection lot will be sharing with OQC. IPQC can be done in multiple activity numbers.

Commented [HH9R8]: Only Lims workflow is described here, if needs SAP integration workflow, we will add it here.

Commented [XW10R8]: 1)IPQC workflow diagram is added.
2) one activity No is corresponding one sample in LIMS, be it IPQC or OQC. SAP will indicate clearly the activity No are for which inspection types.

3) one exception: if IPQC result needs to copy to OQC. Then offline, there is only one physical sample. Meant for IPQC and OQC purpose.
However, in SAP, the inspection lot will have two activity NO for IPQC and OQC respectively.

Commented [XW11]: Resample triggering process:

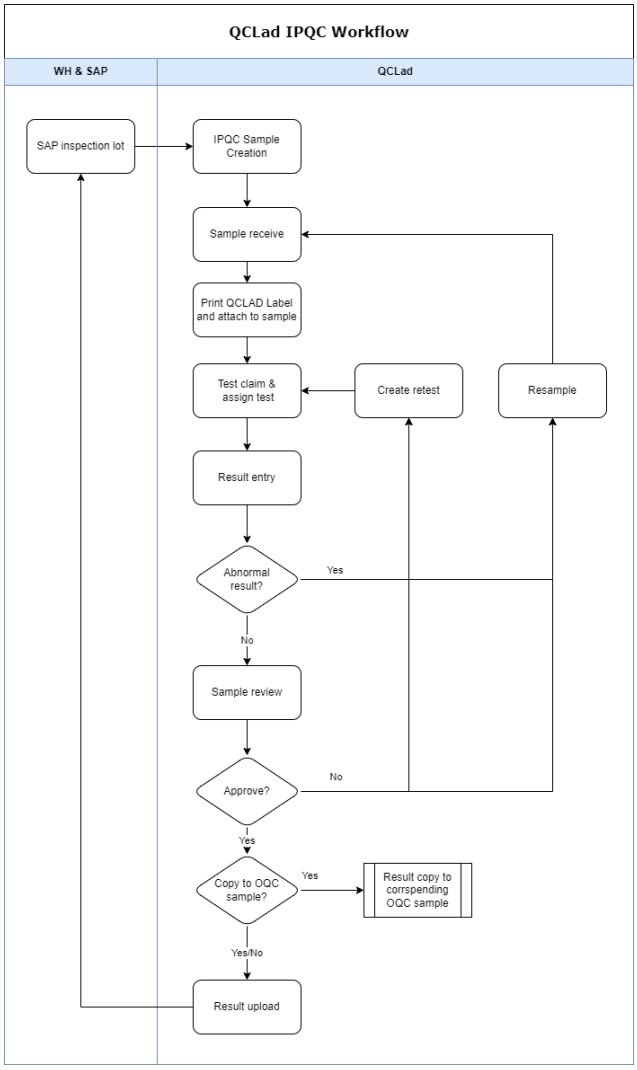
Case 1: before sample review (during inspection, or during sample review)-->**just trigger resample to create another sample but not cancel original sample**-- then need to process these two samples respectively on their own-- the result of both sample should be pushed to SAP.

Case 2: before sample review (during inspection, or during sample review)-->**just trigger resample to create another sample but CANCEL original sample manually don e in LIMS by user**-- then need to process tnew sample—then push to sAP after reveiwe..

Case 3: after sample review, won't allow resample action.

Case 4: sample amount is not enough for retain after sampel review. It has nothing to do with resmaple. Resample is the concept to ask for additional sample for further additional testing but not for retain.

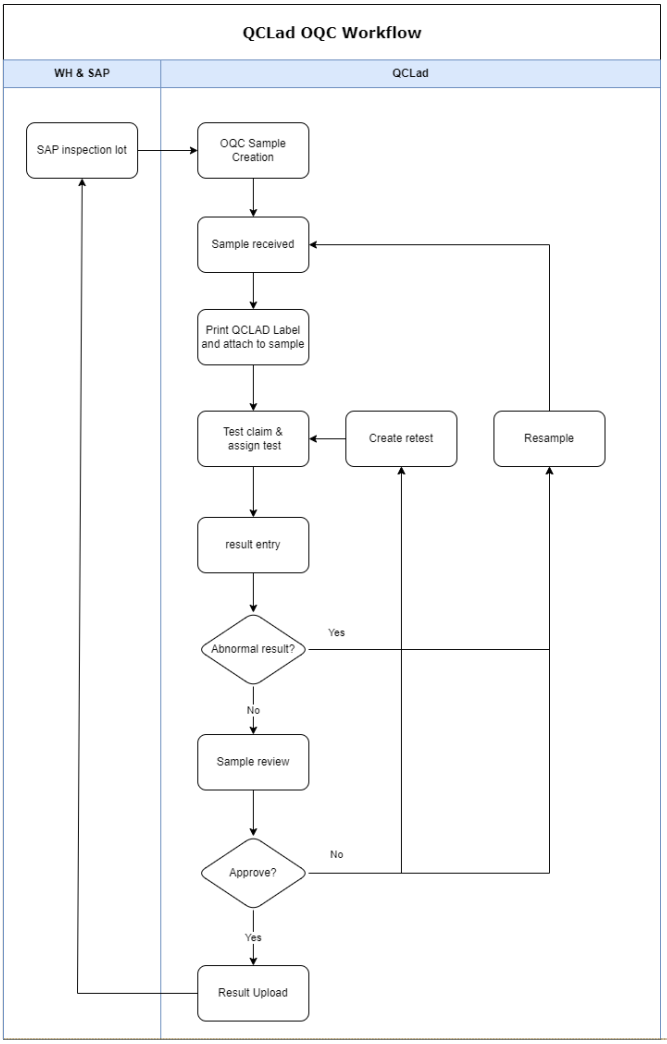
In terms of the process to create the retain sample, whether munaylly or auto, will be discussed further on PPB.



Workflow3: OQC

According to user requirements, for the OQC workflow, we need to add the function of copying IPQC last phase result to OQC as initial sample result, therefore based on main workflow, OQC add some workflow as shown below.

Commented [FS12]: Second checkbox should be Whether IPQC final result copied to OQC



Name	Workflow versi...	Table name	Workflow type	Description	Approval status
OQC	1	SAMPLE	Sample Login	HENKEL	Authorized

Workflow4: Non-standard Testing

According to user requirements, for Non-standard testing workflow, due to external requestor need to raise sample testing via email, finial solution is determined to edit in the excel of the email attachment, therefore Non-standard testing workflow as shown below to meet requirements.

Commented [XW13R12]: Updated in the workflow. Whther copy IPQC results to OQC , this checkbox is added in IPQC workflow diagram.

Commented [NR14R12]: IPQC, OQC inspection lot can be created for material type='UNF' and 'FER'. No sample label is available from MES for OQC, so the flow has to be changed

Commented [XW15]: Current case in Henkel: at the time the external requestor preparing the excel file, the requestor will **indicate in the file that using retain sample for testing or not**.

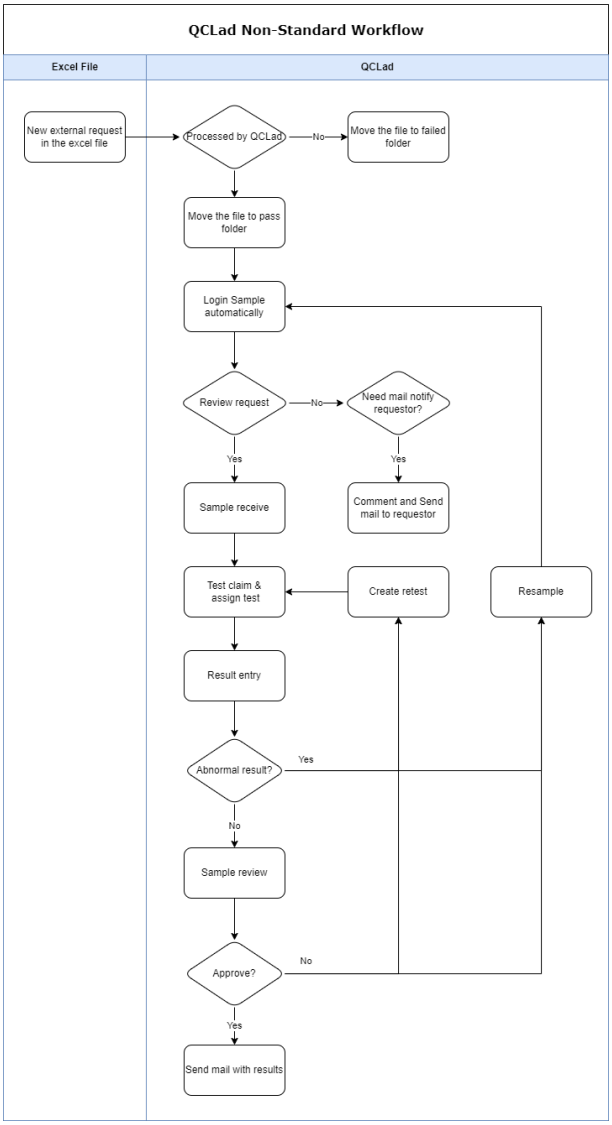
Henkel action item: Henkel should internally align with global team themselves. But for now, go with Dragon's cases.

In LIMS, when processing the file:
In PPA, will simply record this indicator as it is as a sample attribute. However it won't have any impact on the workflow process itself.
And if given as Yes, user should manually key in something in "comment" fields.

And if given as No, user just proceed as it is.

In PPB: will discuss further how the indicator in the excel file impact on the workflow.

QCLad supports to add master data of MLP (analysis and specifications) that are sync from SAP insepction plan.



Name	Workflow versi...	Table name	Workflow type	Description	Approval status
Non-standard testing	1	SAMPLE	Sample Login	HENKEL	Authorized

3.2.1.2 Sample Login

3.2.1.2.1 Business Scenarios

Generally, LIMS should allow user to manually log in different inspection type of samples. And for different inspection type, user can dynamically maintain the sample inspection request template, as of sample entity template in LIMS and its workflow respectively. And user can manually run the workflow and log in the sample as needed.

3.2.1.2.2 RTM Mapping

URS Ref.	Requirement	Type
4.1.1.1.1	System shall be able to login samples using configurable templates	Configuration
4.1.1.1.3	System shall be able to create samples manually for standard testing.	Configuration
4.1.1.1.4	System shall be able to create samples manually for non-standard testing.	Configuration
4.1.1.3.1	System shall assign a unique sample ID to each sample logged in.	Configuration
4.1.1.6.1	System shall be able to define configurable attributes for a sample which can be modified.	Standard
4.1.1.7.1	System shall be able to allow the definition of sample types and test schedule.	Standard
4.1.1.7.3	System shall be able to log in predefined batches of samples at one time in accordance with a defined sampling plan.	Configuration
4.1.1.8.1	System shall be able to log in either single or multiple samples from the same batch.	Standard
4.1.1.10.1	System shall be able to log in non routine samples.	Standard
4.1.1.10.2	System shall allow specific user to select specifications and test methods for non routine samples.	Standard
4.1.1.15.1	System shall be able to allow user to enter comments at Sample log-in.	Configuration

3.2.1.2.3 Function Design

Requirements Description :

Users can create samples and tests with unique ID for routine and non-routine testing through configurable templates, and users can add comments while sample login manually. (Based on RTM as above)

Design Overviews :

a. Table change

Add new table

i: table_plant

ii: table_ work center

Add new fields in sample table

i: idh, text

ii: plant code, link to table plant

iii: storage condition, text

iv: activity no, real

v: work center, link to table work center

vi: mfg date, date

vii: expired date, date

viii: test item, text

ix: requestor, text

x: requestor address, text

b. Entity templates

Add four configurable entity templates for four different workflows in Phase A.

i: IQC entity templates, description: For Standard RM

ii: IPQC entity templates, description: For without MES

iii: OQC entity templates

iv: Non-standard testing entity templates

c. Workflows

Add four sample login workflows for creating new samples and tests.

i: IQC workflow, description: For Standard RM

ii: IPQC workflow, description: For without MES

iii: OQC workflow

iv: Non-standard testing workflow

d. Cabinet and folders

Maintain cabinet and folder for sample management.

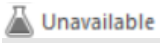
i: add sample management cabinet

ii: add sample login folder

iii: add new sample pending receiving folder

e. Functional logic

While the sample login successfully, the sample automatically created and automatically record the current server time and operator:

i: The sample status update as "unavailable" 

ii: The operator is recorded as "created by"

iii: The sample created exist in sample pending receiving folder.

Labels can be printed manually in sample pending receiving folder.

Function Entrance :

Sample Management Cabinet →Sample Login Folder→Right click menu "Run Workflow"

Commented [NR16]: Can you please provide which will be pulled from label, SAP and QCLAD master data?

Commented [MX17R16]: We will explain this on Thursday

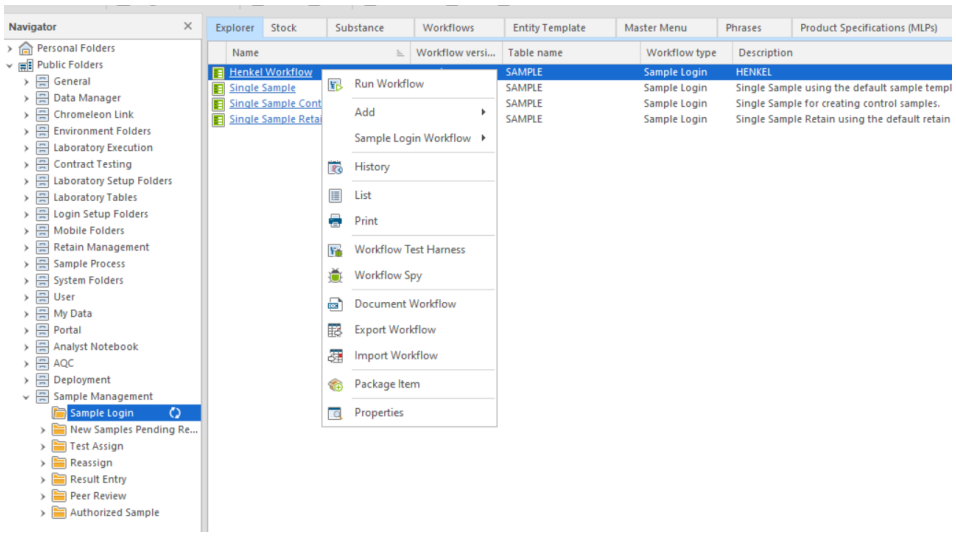
Commented [NR18]: This is retain storage condition and it needs to be maintained as master data in QCLAD

Commented [CG19R18]: Plant code will in master data.

Commented [NR20]: Is this batch mfg and expired data?

Commented [MX21R20]: Yes

User can start manual login samples by right click 'run workflow' button on specific workflows. Sample login interface will be different according to requirements based on different sample type.

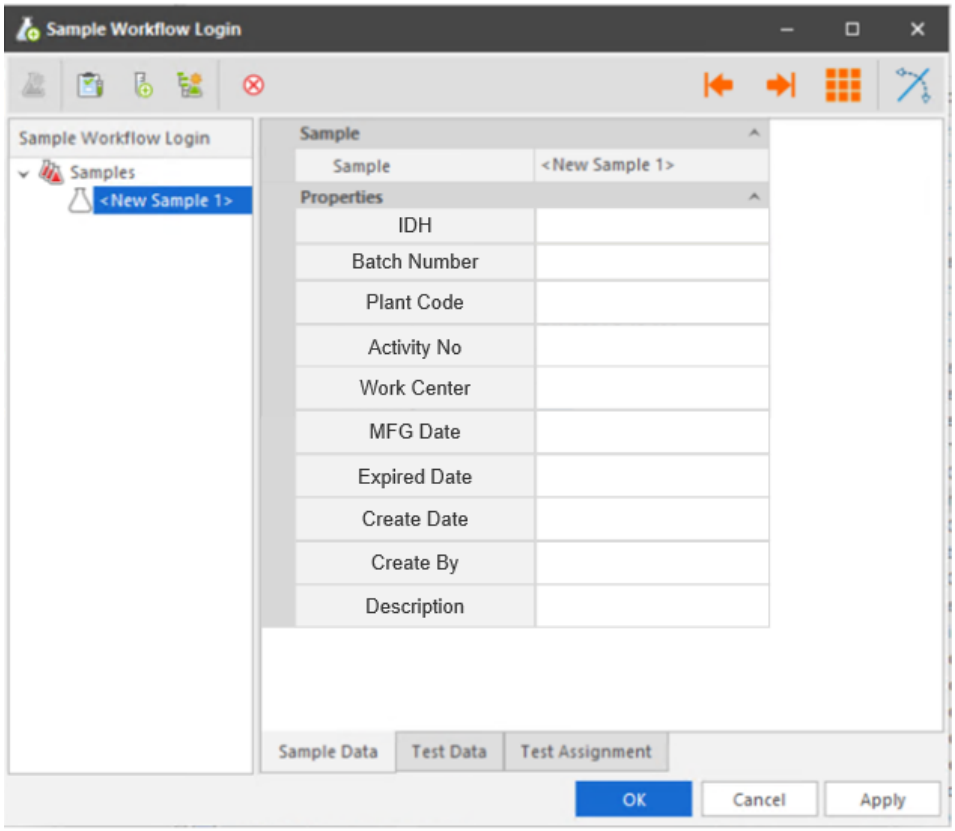


User Interface Design Reference :

According 3.2.1.1 and requirements, LIMS will create four configurable entity templates for four workflows in Phase A. Detail properties and user interface reference as below:

Note: According to the *requirements description*, this part of the content for creating samples manually, the data fields corresponding of samples pulled from SAP will be detailed in Part xx.xx.xx, see the Interface module for more information, so will not go into the details of the fields and SAP associated here.

Entity template 1: for IQC (Standard RM) workflow

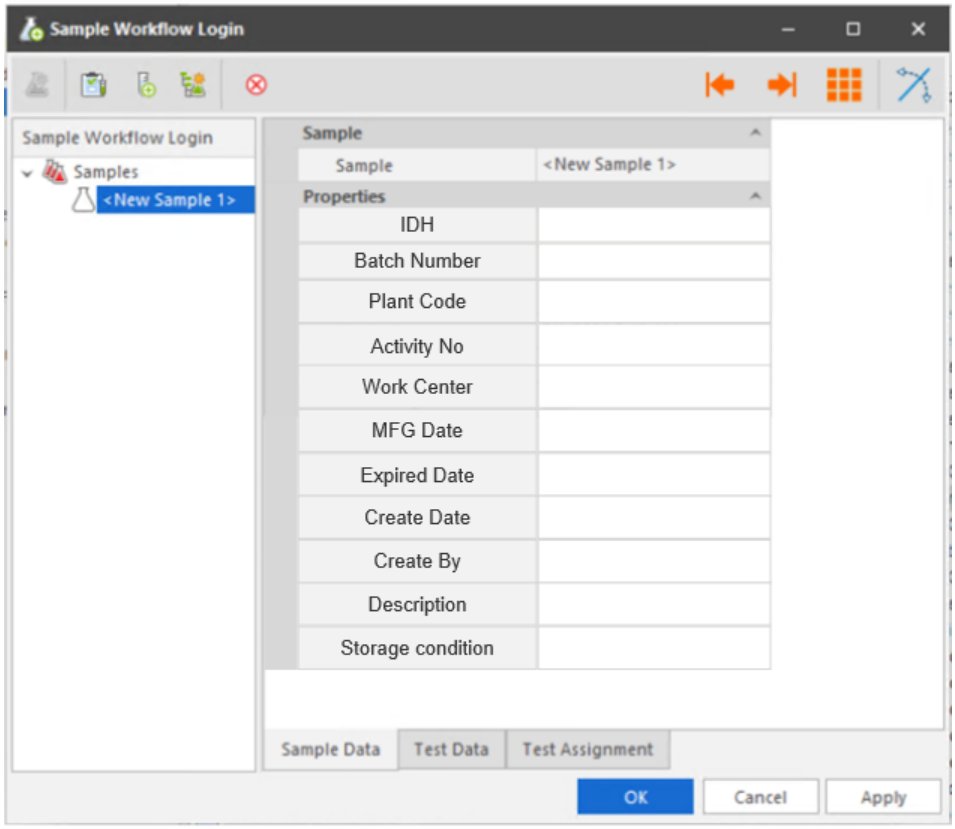


Properties in entity template clarification in LIMS:

No.	Properties	Type	Prompt Type	Description
1	IDH	Numeric	Mandatory	Links to MLP table
2	Batch	Text	Mandatory	Batch id
3	Plant code	Text	Optional	Links to Plant table
4	Activity No	Numeric	Optional	Activity number
5	Work Center	Text	Optional	Links to Work centre table
6	Create Date	Date	Optional	Current date, modifiable.
7	Create By	Operator	Optional	Current operator

8	Description	Text	Optional	Descriptiom
9	Sample Type	Default	Hidden	IQC； sample type workflow is automatically assigned by lims
10	MFG Date	Date	Optional	Manufacture date
11	Expired Date	Date	Optional	Expiration date
14	Test Schedule	Default	Hidden	

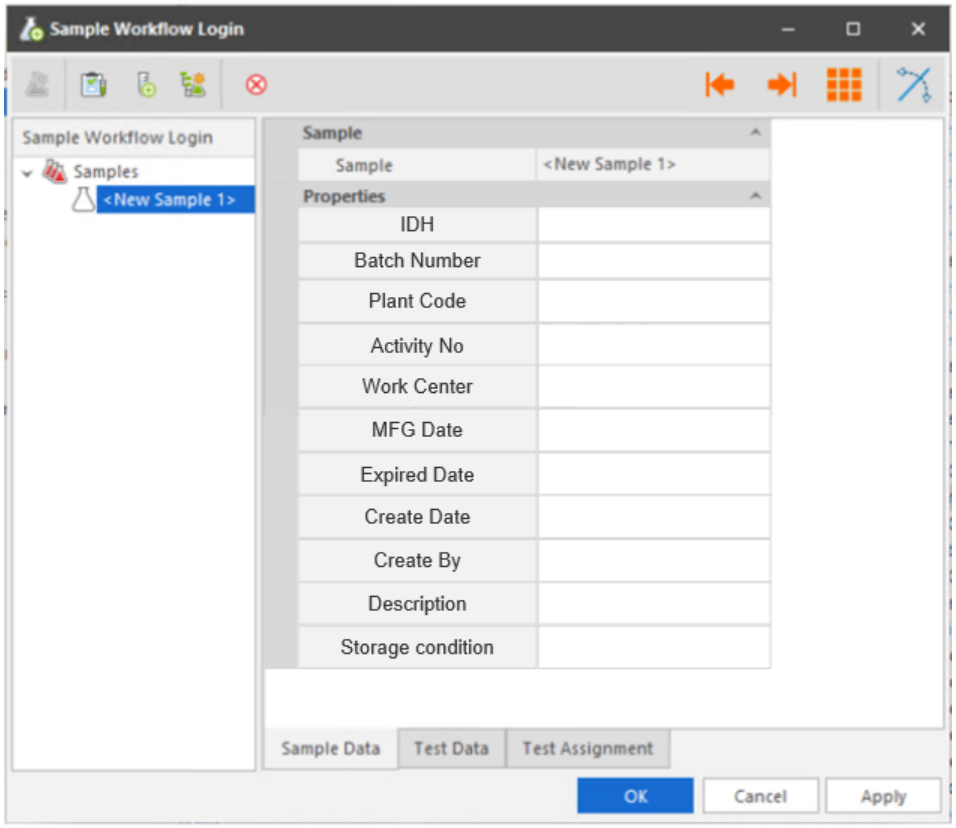
Entity template 2: for IPQC (without MES) workflow



Properties in entity template clarification in LIMS:

No.	Properties	Type	Prompt Type	Description
1	IDH	Numeric	Mandatory	Links to MLP table
2	Batch	Text	Mandatory	Batch id
3	Plant code	Text	Optional	Links to Plant table
4	Activity No	Numeric	Optional	Activity number
5	Storage Condition	Text	Optional	Fetch from substance table
6	Work Center	Text	Optional	Links to Work centre table
7	Create Date	Date	Optional	Current date, modifiable.
8	Create By	Operator	Optional	Current operator
9	Description	Text	Optional	
10	Sample Type	Default	Hidden	IPQC; sample type workflow is automatically assigned by lims
11	MFG Date	Date	Optional	Manufacture date
12	Expired Date	Date	Optional	Expiration date

Entity template 3: for OQC workflow



Properties in entity template clarification in LIMS:

No.	Properties	Type	Prompt Type	Description
1	IDH	Numeric	Mandatory	Links to MLP table
2	Batch	Text	Mandatory	Batch id
3	Plant code	Text	Optional	Links to Plant table
4	Activity No	Numeric	Optional	Activity number
5	Storage Condition	Text	Optional	Fetch from substance table
6	Work Center	Text	Optional	Links to Work centre table

7	Create Date	Date	Optional	Current date, modifiable.
8	Create By	Operator	Optional	Current operator
9	Description	Text	Optional	Description
10	Sample Type	Default	Hidden	OQC; sample type workflow is automatically assigned by lims
11	MFG Date	Date	Optional	Manufacture date
12	Expired Date	Date	Optional	Expiration date

Entity template 4: for Non-Standard Testing workflow

Commented [FS22]: Complete date of the request and whether using retain sample shall be added

Commented [MX23R22]: Need to discussion on Thursday

Commented [CG24R22]: Expected completion date, display for lab user reference, in non-std testing

Excel fields	Mapping in QCLAD application（Description）
Requestor's Department	Display
Name of Requestor	Display
Email Address	Display and to send back results back
Region	Display
Country	Display
Plant Name	Display
Plant code	Need to be mapped with SAP plant code
Work center	For those IDH with inspection plan QC LaD need to pull the work center from inspection plan, considering IDH & plant code. For "work center" entry by requestor, QC LaD shall only display those comment with no interface required.
IDH	Need to be mapped with SAP plant code to pull the right inspection plan
Material Description	Display
Batch Number	Display

Test	Display. QC operator will manually read the Test and map it with the right MIC of SAP inspection plan. QCLAD only needs to display the inspection plan
Use retain sample or not	To check whether retain sample is required in the non standard testing workflow
CCMS, SCMS, NCMR Number(If Any)	Display
Expected completion Date	Display

Ground Rules

1. For material within SAP, system shall sync & pull the work center
2. For material not in SAP, a dedicated QC LaD user shall select the work center from the task pool with the hint from manual entered work center(free text)
3. For CCMS/NCMR/SCMS related test request, work center shall be pulled from SAP(workflow #9)
4. For other test request, work center shall be pulled from SAP for those material have IDH, otherwise test request shall go to the task pool awaiting manual workcenter selection by supervisor/shift leader.(Workflow #11)
5. Requestor can raise multiple test requests at once by simply add additional roles

	No.	Requestor's Departaent	Name of Requestor	Email Address	Region	Country	Plant Name	Plant code	Work center	Material IDH	Material Description	Batch Number	Test	Use retain sample or not	CCMS, SCMS, NCMR Number(If Any)	Expected completion Date
Sample	1	Quality	David Li	david.li@henkel.com	APAC	China	Wai Gao Qiao	3372		1234567	84-1A	C84D001234	Test 1: Viscosity test @5rpm Test 2: Potassium content	Yes	NA	08/31/2024

Excel Sample template

Interaction Events:

Click the "Apply" button to complete sample login operation. The system will automatically create the samples and tests as shown below.

Commented [FS25]: Workcenter

Commented [MX26R25]: Ok

Commented [DL27R25]: Please park "workcenter" for internal discussion.

Commented [CG28R25]: tbd

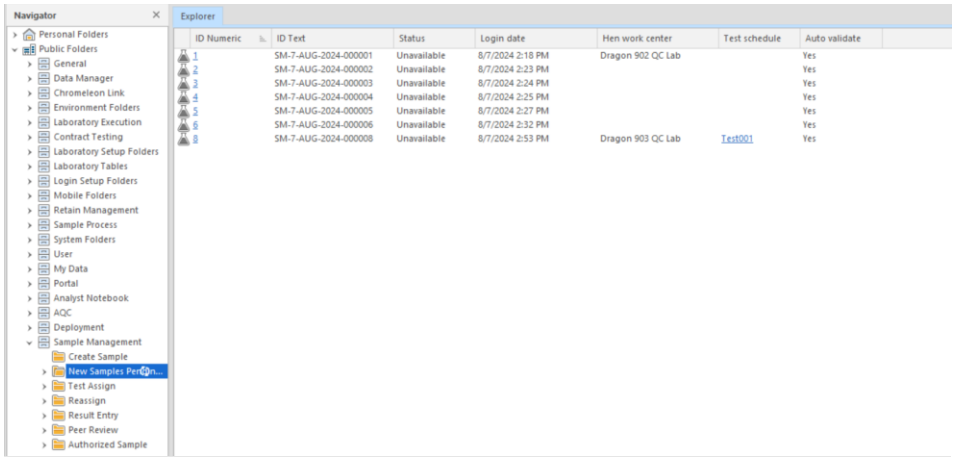
Commented [NR29]: Is this email address, requestor's email address?

Commented [MX30R29]: yes

Commented [FS31]: Need to add whether use retain sample or not and CCMS,SCMS, NCMR number

Commented [XW32R31]: Needs to update the screenshots of the updated Excel templates.

Action item: Frank will provide the new excel template later.



Situation clarification:

Sample Login three ways

- ***Automatically***: LIMS create samples and tests automatically based on inspection lot by SAP interface. Inculding: IQC(standard RM)、IPQC 、OQC workflow. Sample type and material description will be automatically assigned by QCLad.
- ***Manually***: User right click on specific workflow, click 'run workflow' RMB button and enter mandatory properties to create samples and tests. For example: temporarily testing.
- ***Special***: For Non-Standard entity template, requestor needs to fill specific excel table in shared disk pathway regarding user requirements. LIMS will create samples without test items based on excel items. Inculding: Non-Standard testing workflow. Sample type and material description will be automatically assigned by QCLad.

Commented [NR33]: What are the criteria for this?

Commented [CG34R33]: Virtual sample creation in lims. Add description sample type workflow is automatically assigned by lims

3.2.1.3 Sample receiving

3.2.1.3.1 Business Scenarios

QCLad allows you to perform certain processing actions with a barcode scanner. Multiple samples can be scanned sequentially to complete an action, thereby multiple samples can be received at the same time by scanning label with MES barcode, and then user can print LIMS label within QCLad manually.

3.2.1.3.2 RTM Mapping

URS Ref.	Requirement	Type
4.1.1.3.2	System shall be able to scan the Labels from MES system and fetch unique ID.	Customisation

4.1.1.3.4	System shall allow user to print QCLaD Labels if the Plant doesn't use MES system.	Standard
4.1.1.4.1	System shall be able to retrive the sample receive date and time.	Standard
4.1.1.16.1	System shall be able to scan barcode or QRCode and match barcode or QRCode sample in QCLAD system.	Customisation
4.1.1.18.1	Sytem shall be able to allow entry / modification of attribute data at point of single or multiple sample receipt using configurable templates.	Configuration
4.1.1.18.2	System shall be able to receive sample by scanning MES label	Customisation
4.1.1.18.3	System shall be able to create labels for sample scan receiving for non MES samples.	Configuration

3.2.1.3.3 Function Design

Requirements Description :

Users can receive samples with scaning the Labels from MES system and entry/modify some modifiable fields while receiving sample. After completing receiving sample, user can print sample label within QCLad. (Based on RTM as above)

Design Overviews :

a. **Table change**
   Add new fields in sample table
   - i: hen_mes_id, text
   - ii: hen_received date, date
   - ii: hen_received by, operator

b. **Barcode Process Layout**
   Modify Columns of Sample Receive Barcode Layout:
   - i: hen_mesid
   - ii: hen_idh
   - iii: batch number
   - iv: hen_workcenter
   - v: idtext
   - vi: hen_received date
   - vii: hen_received by

c. **Folders and mater menu**
   Maintain menu in new sample pending receiving folder.
   Add Test assign folder.
   Add Samples in Testing folder.

d. **Functional logic**
   While the sample is received, the tests automatically created based on test schedule and automatically record the current server time and operator（in folder and dashboard, these time will displayed by client time zone）：

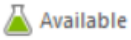
Commented [NR35]: From where are we pulling workcenter?

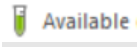
Commented [NR36]: what do you mean received completely? Is this step happened after physical sample is actually available. Is there any scenario, were the physical sample received but Sample not in Available state?

Commented [MX37R36]: Misunderstanding, we will delete the completely

Commented [NR38]: Please ensure we show the date and time based on the plant time zone.

Commented [MX39R38]: Add explain

i: The sample status update as "Available" 

ii: The operator is recorded as "received by"
iii: The current date is recorded as "received date"
iv: The tests status update as "Available" 

v: The tests created exist in Test assign folder.

MES Labels can be scanned using receive sample interface of master menu 31026 in sample pending receiving folder and match corresponding fields.

Detail action description for workflow：

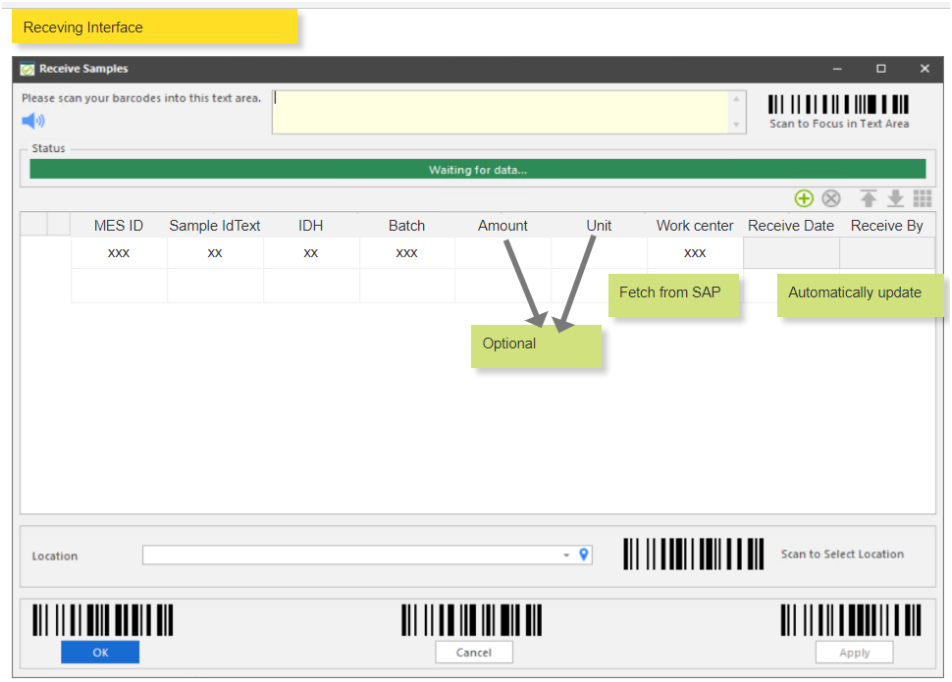
- For workflow IQC (Standard RM)：
  - Sample receiving by scanning MES labels.
- For workflow IPQC(without MES):
  - Sample receiving by manually click "Receive Sample" menu.
- For workflow OQC：
  - Sample receiving by manually click "Receive Sample" menu.
- For workflow Non-Standard Testing：
  - Sample receiving by manually click "Receive Sample" menu.

Commented [NR40]: MES labels are only available for IQC, IPQC(with MES). IPQC(with MES) and OQC labels have to be created within QCLAD

Commented [MX41R40]: Sure, will add this

Function Entrance：

Sample Management Cabinet → New sample pending receiving Folder→Right click menu "Receive Sample", then will pop up receive sample window as shown here.



Sample receiving can also be operated by scanning MES labels, clicking "Receive Sample" menu in the menu bar above SampleManager, the sample receiving interface can also be popped up;

Users can scan the MES label in this interface, the system will automatically retrieve the unique sample displayed in the interface, you can scan more than once;

MES tag format: { Plant; material; batch; JIT or not; MES Sample ID; sampling date; sampling time; inspection type }

Example:

{'0759';'2870306';'45';'NOT';'0000116885';'2024-06-11';'04:11:10 PM'; 'IQC'}

{'0759';'2870306';'45';'JIT';'0000116885';'2024-06-11';'04:11:10 PM'; 'IQC'}

The system can retrieve the unique sample by scanning out the Plant, material, batch, inspection type and JIT or not, MES Sample ID, sampling date (including sampling date and sampling time) is automatically written into the sample and displayed in the interface;

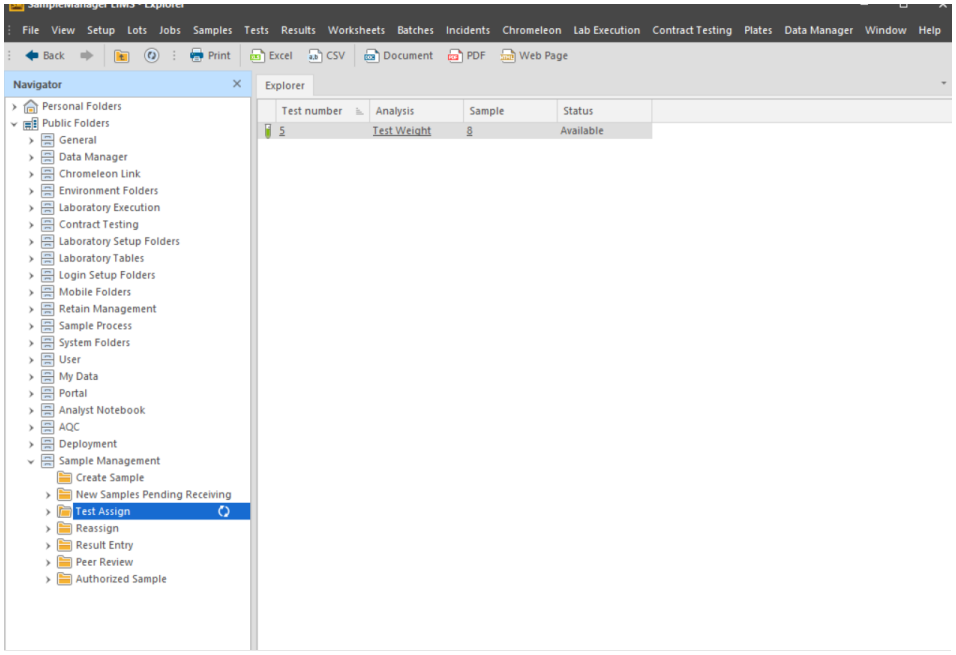
Interaction Events:

- ***Receiving sample***

  Click the “Apply” button to complete sample receive operation.

The system will automatically close the window and create the tests in Test assign folder as shown here.

And the sample status will change to “Available” and will not shown in New sample pending receiving Folder anymore.



Note: The columns on the Explorer can be configured according to the display needs.

- ***Print sample label in QCLad***

In samples in testing folder, user can select specific sample and then right click the “Print Label by Template” button to execute print sample label operation.

QCLad supports 2 way of label printing:

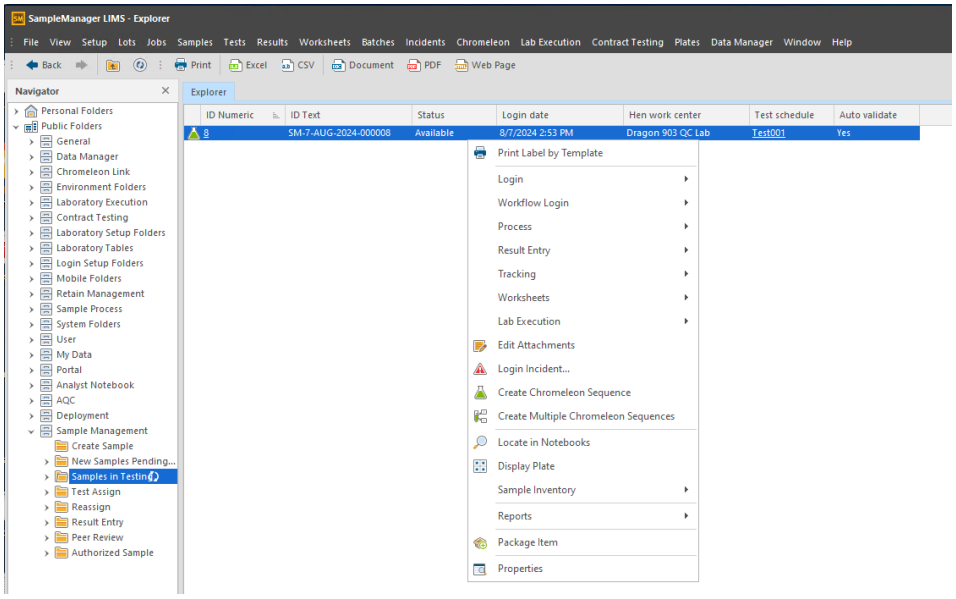
1. Local printer printing label: will use client computer’s printer
2. Network printing label: will use the printer installed on the lab server. Only within this way QCLad can print label automatically when sample is received.

Commented [FS42]: QCLaD should automatically print the label once the sample is received.

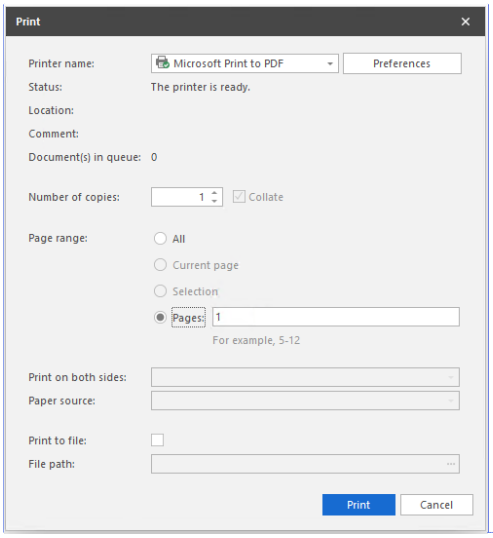
Commented [MX43R42]: We will discussion this

Commented [CG44R42]: 1.one printer in one lab. 2. network printer to be connected and driver installed in qclad server

Commented [DL45R42]: Please update the FRS with the function to automatically print the label right after the sample receiving accordingly



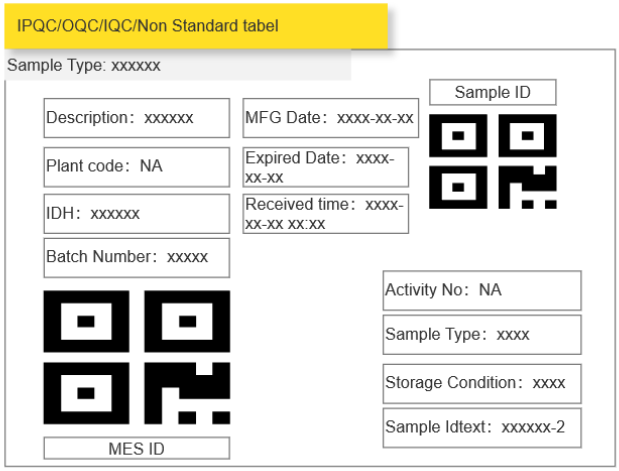
The system will automatically pop up the window and print as shown here. Network printing will not use this window

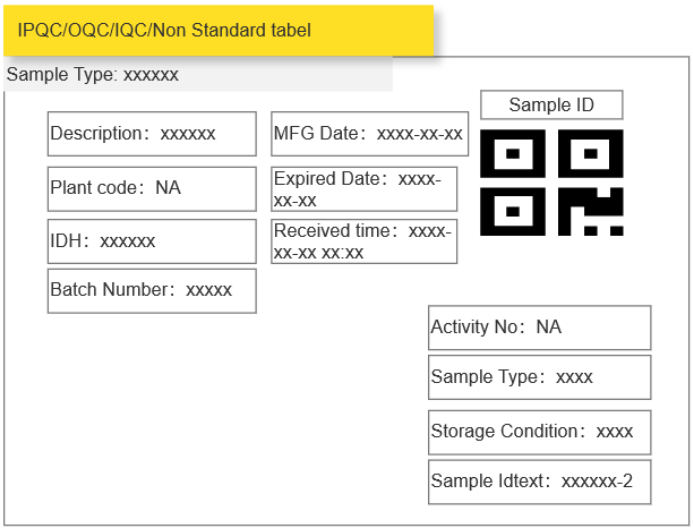


Commented [FS46]: We don't expect this interface, workload will be increased.

Commented [CG47R46]: Screenshot is local printer printing. We can provide 2 way label printing. Network printing will not use this window

Note: label template prototype example as shown below:





Description in detail:

Sample type	depends on the defined workflow	IQC\IPQC\OQC\NSQC
Sample ID	Generate from the QCLad	

Commented [FS48]: For sample with MES label, we only need to print sample ID

Commented [CG49R48]: Label template. Need to fit paper size. Stick on top of mes label. Contents on the label will be decided later. 3 label size (std, small, large)

Commented [FS50]: Sample without QCLaD's fields origin will be different from that of sample with QCLaD

Commented [CG51R50]: Sample id is lims numeric id

MES ID	fetch from the MES Label	
Plant code	Fetch from the SAP	
IDH	Fetch from the SAP	
Batch Number	Fetch from the SAP	
MFG Date	Fetch from the SAP	
Expired Date	Fetch from the SAP	
Received Date	Generate from the QCLad	
Activity No	Fetch from the SAP	
Storage Condition	From QCLad master data.	User can manual enter in IDH master data if needed.
Sample Idtext	Generate from the QCLad	
Description	Fetch from the SAP	IDH description

Commented [NR52]: Material IDH and batch number will be received from SAP as well as from MES label if there is one.

Commented [MX53R52]: We only use Sap

Commented [NR54R52]: Ok, please ensure we have to compare this information with MES to get the right sample.

Commented [CG55]: From lims master data. User can manual enter in IDH master data if needed.

Commented [NR56]: what is this description?

Commented [MX57R56]: Dragon want add this

Commented [NR58R56]: Is this material description? Please be specific.

Commented [CG59R56]: IDH description

3.2.1.4 Tests Preparation

3.2.1.4.1 Business Scenarios

Some analysis requires preparation of the specimen prior to performing the test. Users can specify the pre-test preparation process required for a particular analysis.

3.2.1.4.2 RTM Mapping

URS Ref.	Requirement	Type
4.2.1.1.1	System shall be able to have option for a sample preparation which may be required for some analysis.	Standard
4.2.1.1.2	System shall be able to assign preparation to an operator.	Customisation

3.2.1.4.3 Function Design

Requirements Description :

Many laboratory processes entail some form of preparation, for example filtering samples that contain particles or diluting concentrated samples. These can be important steps in analytical terms and so must be handled by SampleManager. Preparations are static data items that can control the handling of samples and tests. Preparations linked to samples and / or tests can affect the status values of both. Preparations are entered into the system via a typical Data Entry screen. The Wait for Preparation field can be set to Yes or No. If set to Yes then the related sample or

tests status becomes W. When set to Yes then a Sample or Test will require positive action to complete the preparation and progress through the system can only take place after this positive action has been taken. Samples and Tests with Preparations assigned can be handled in bulk through the relevant Preparation Worksheet. Preparations may be found in the Environment Folders / Preparations folder.

Design Overviews :

a. **Table change**
   Add new fields in test table, to record preparation start date, target date and complete date.
   Add new fields in test table, to record prepartion assign person.
   Add new fields in test table, to record prepartion start person and preparation complete person.
b. **Workflow**
   Add nodes in test lifecycle:
   1. Start preparation: If the test has preparation method, then mark the test is in preparation status, set start date and start person. Automatically calculate the target date based on analysis static data.
   2. Complete preparation: if time now is larger than target date, mark the test is completed preparation. Allow the test to continue further process.
c. **Folders and Cabinets**
   Multiple folders will be added into sample management cabinet for usability:
   1. Tests waiting preparation: tests that has preparation step and its sample has been received.
   2. Test in preparation : tests are in preparation step currently.

Interaction Events:

Four workflows in Phase A are all adapt to this operation, therefore will not description in detail here.

- ***Static Data maintenance***

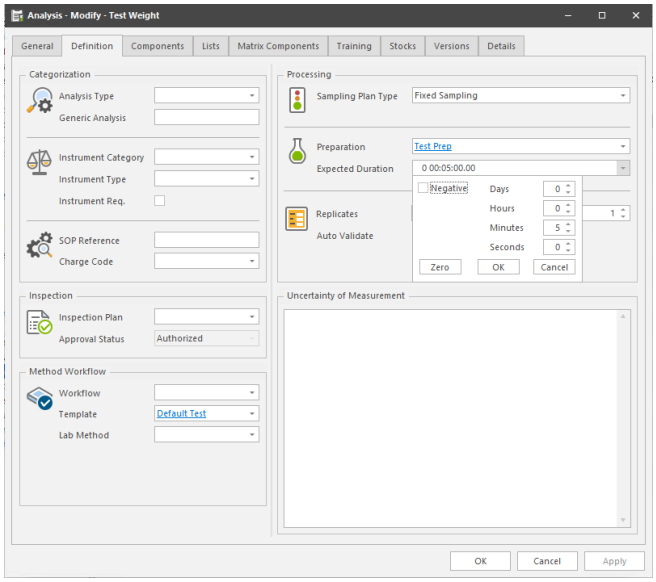
  Maintain and add preparation methods in certain analysis.

  Test preparation on mic level. QCLad can distinguish samples which are in progress or not started. Display hen_status for user understanding in sample by adding new field in structure of QCLad.

Commented [NR60]: Is preparation master data at MIC level or inspection lot level?

Commented [MX61R60]: MIC level

Commented [CG62R60]: Test prep on mic level. Lims can distinguish samples which are in progress or not started. Display henkel status for user understanding in sample.

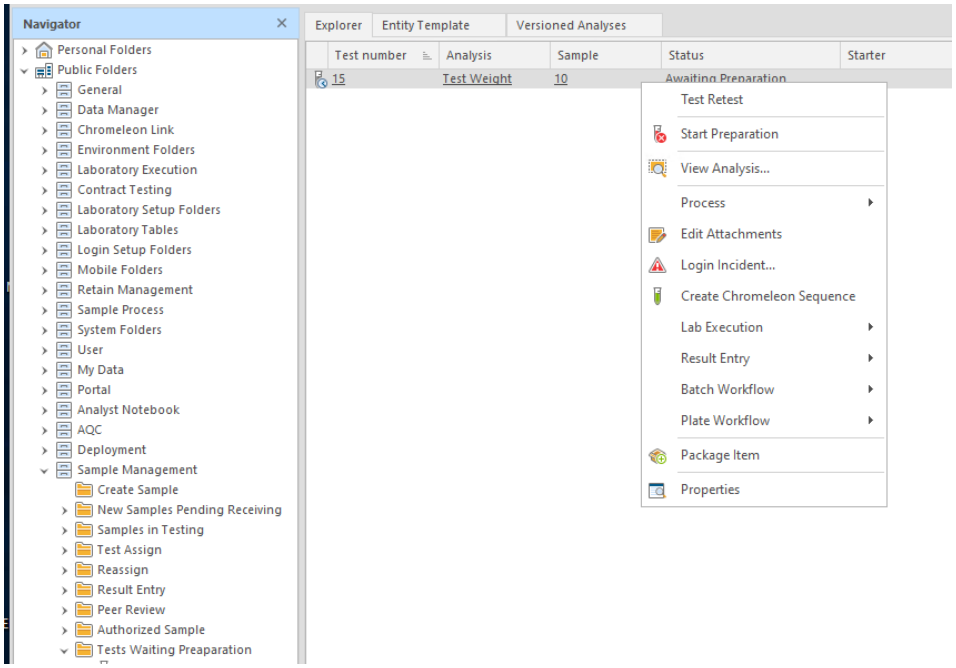


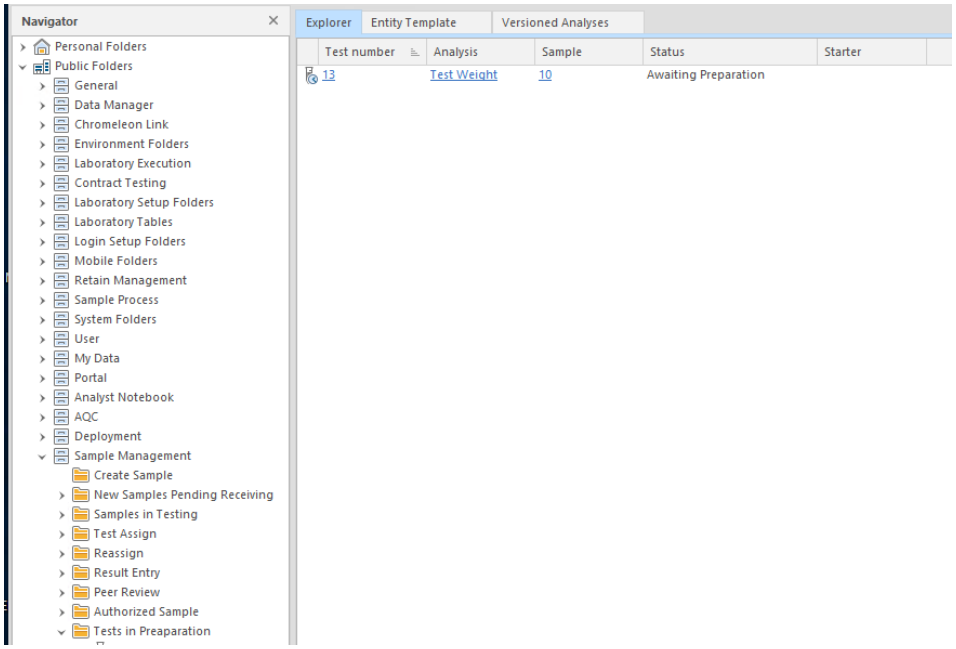
- ***Preparations***

When a new sample that has test need to be prepared get received in lab, the certain test will enter awaiting preparation status in tests waiting preparation folder. Test will enter waiting preparation folder. Sample or tests can appear in multiple folders. User can start prepartion by RMB button 'start preparation'. system will calculate the target completation time based on the analysis setup.

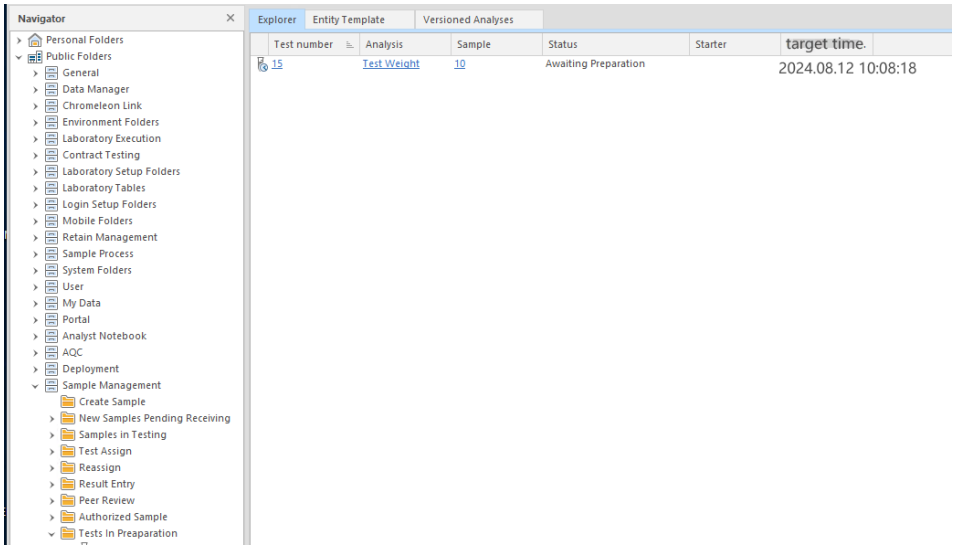
Commented [FS63]: In which situation will sample enter waiting preparation folder?
Can a batch of sample appear in mutiple folder?

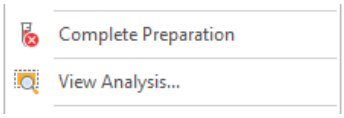
Commented [CG64R63]: Test will enter waiting preparation folder. Sample or tests can appear in multiple folders


Navigator
Personal Folders
Public Folders
General
Data Manager
Chromeleon Link
Environment Folders
Laboratory Execution
Contract Testing
Laboratory Setup Folders
Laboratory Tables
Login Setup Folders
Mobile Folders
Retain Management
Sample Process
System Folders
User
My Data
Portal
Analyst Notebook
AQC
Deployment
Sample Management
Create Sample
New Samples Pending Receiving
Samples in Testing
Test Assign
Reassign
Result Entry
Peer Review
Authorized Sample
Tests Waiting Preaparation
Explorer
Entity Template
Versioned Analyses
Test number
Analysis
Sample
Status
Starter
15
Test Weight
10
Awaiting Preparation
Test Retest
Start Preparation
View Analysis...
Process
Edit Attachments
Login Incident...
Create Chromeleon Sequence
Lab Execution
Result Entry
Batch Workflow
Plate Workflow
Package Item
Properties



After user click start preparation, User can complete the preparation only when time is arrived. System will not allow user to enter results without complete preparation.



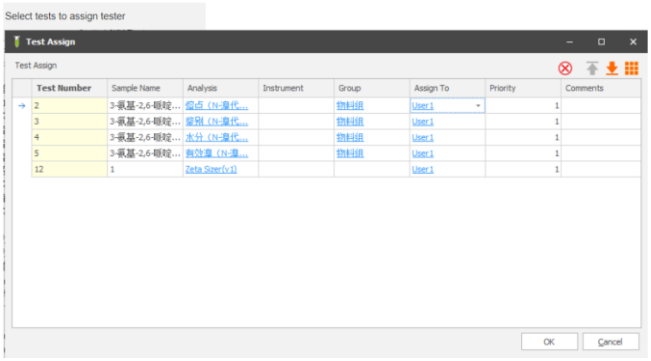


After the test is completed preparation, it can contine with further process.

Function Entrance：

Sample Management Cabinet → Tests waiting preparation Folder→Select single or multiple tests and right click menu "Claim test Preparation", then the test items will disappear immediately under Test Assign folder and shown in Result entry folder.

Sample Management Cabinet → Tests waiting preparation Folder→Select single or multiple tests and right click menu "Tests Preparation Assign", then will pop up window as shown here and user can select specific operator to execute the test.



The people filtering filters are as follows：

Manual assign- selecting operator Filters:

1. lab(work center).
2. On-duty personnel (shift management).

User can assign test operator by multi select tests.

Commented [ZZ65]: People filtering filters to be discussed and confirmed

Commented [CG66R65]: Manual assign- selecting operator Filters: based on 1. lab(work center). 2. On-duty personnel (shift management).
User can assign test operator by multi select tests.
Test preparation operator should be same as test entry operator. Only need to assign once. Assignment can rearrange to another user.

3.2.1.5 Tests Assign

3.2.1.5.1 Business Scenarios

When sample is received to lab, test items need to be assigned to lab operators so that lab operators can view their task list and start their own work. This assignment can be done automatically based on fixed rules or by manually assign. The assigned tests can also be reassigned to others.

3.2.1.5.2 RTM Mapping

URS Ref.	Requirement	Type
4.1.2.1.1	System shall be able to assign tests to an operator manually or automatically.	Customisation
4.1.2.1.3	System shall support to stop or modify auto assign task.	Customisation
4.2.1.1.6	System shall be able to have standard time defined for each test assignment. This shall come from the master data to LIMS.	Configuration
4.2.1.1.16	System shall be able to assign samples manually.	Customisation
4.2.1.8.1	System shall allow user maintain master data for shift schedule by week. Number of shift and shift hour of each shift shall be configurable in the master data at plant level.	Customisation
4.2.1.9.3	System shall allow user to rearrange task manually.	Configuration

3.2.1.5.3 Function Design

Requirements Description :

Users can manually assign tests to an operator or claim tests by themselves. And system support assign tests automatically based on shift rule defined in QCLad.

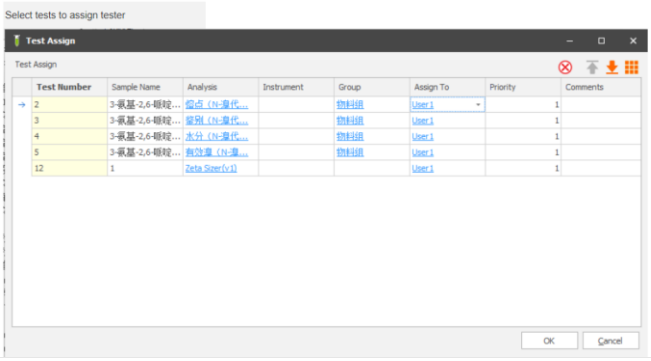
Design Overviews :

a. **Table change**
   Add new fields in test table, to record testing assign person.
b. **Folders and mater menu**
   Add Samples in Testing folder.
   Add Reassign folder
c. **Functional logic**
   While the test is assigned to an operator completely:
   i: The starter of test update as assigned user.
   ii: The tests automatically existed in Reassign folder

Function Entrance :

Sample Management Cabinet → Test Assign Folder→Select single or multiple tests and right click menu “Claim test”, then the test items will disappear immediately under Test Assign folder and shown in Result entry folder.

Sample Management Cabinet → Test Assign Folder→Select single or multiple tests and right click menu “Test Assign”, then will pop up window as shown here and user can select specific operator to execute the test.



The people filtering filters are as follows:

Manual assign- selecting operator Filters:

1. lab(work center).
2. On-duty personnel (shift management).

User can assign test operator by multi select tests.

Interaction Events:

Note: Four workflows in Phase A are all adapt to this operation, therefore will not description in detail here.

- ***Assign tests***

  System has three ways to assign tests to technician.

  - User can assign tests to an operator then the technician can entry results for tests.
  - User can claim tests by themselves and then they can entry results for tests.
  - System can automatically assign tests to an operator based on shift rule.

- ***Reassign tests***

  System support reassign tests for some situations, when tester needs to be changed, privileged user can move to Reassign folder to reassign test to another operator.

3.2.1.6 Result Entry

3.2.1.6.1 Business Scenarios

Generally, an analysis must have at least one component defined. A component is an item for which a 'result' * will be expected. System support entry results by single or multiple samples or by single or multiple tests.

Some details will be mandatory (*) for certain data/result types.

Commented [ZZ67]: People filtering filters to be discussed and confirmed

Commented [CG68R67]: See beofre

Commented [DL69]: Please ensure to include peer review & resampling to this chapter which shall be applied to all workflow.

Resampling: resample shall be available within any phase needed in workflow. Henkel aligned all resample data will be pushed to SAP, For some scenario like wrong sample been tested or wrong measurement method used Henkel can use "cancel" or "retest" function to manage. So please make sure to include "cancel" & "retest" mentioned in FRS.

Peer review: Before push data to SAP a "peer review" or "review" is mandatory. Henkel request TF to make sure the peer review function should be configurable in plant level(Go with peer review or all authorized QC LaD user can do the review)

Commented [CG70R69]: Already discussed in previous meeting

Numeric	Min/Max values set the range for acceptable or reasonable results. For example, pH values must lie between 1 and 14.
Boolean	True Word / False Word values will show the result options when a result passes / fails a Boolean test. 'True' & 'False' are the defaults.
Character	Allowed Chars values show the allowable characters when a result is based on a letter or symbol. 'A-Z' are the defaults
Textual	Free text entry.
Option*	Option shows the ID of a Phrase List that has been created as static data. The list has the options that can be selected at result entry.
Calculated* (K)	Calculation or Formula value shows the ID of a Calculation or the Formula definition that has been created as static data.

3.2.1.6.2 RTM Mapping

URS Ref.	Requirement	Type
4.1.1.19.1	The system shall provide functionality to allow entry of any irregularities recorded sample according to laboratory business rules.	Standard
4.1.1.19.2	The system shall provide functionality to allow entry of any routine sample according to laboratory business rules.	Configuration
4.1.2.3.1	System shall be able to display all results data to the sample.	Standard
4.1.2.4.1	System shall be able to calculate results based on reported or recorded data which from instruments.	Configuration
4.1.2.5.1	System shall be able to record all data in whole testing process, including raw data captured from instrument, entry data, data calculated by formular, reported data, data which push to SAP.	Standard
4.1.2.5.2	System shall be able to mark final results against specification limits and flag failures.	Configuration
4.1.2.7.1	System shall be able to support results entry for multiple tests performed on a single sample.	Standard
4.1.2.8.1	System shall be able to support results entry for an individual test on a single sample.	Standard
4.1.29.1	System shall be able to support results entry for single or multiple tests performed on multiple samples.	Standard
4.1.2.10.1	System shall be able to verify entered results data (e.g., pH, sample color, specific gravity) against specification limits.	Standard

4.1.2.11.1	System shall be able to prevent the acceptance of partial or incomplete data entry for results and active samples.	Configuration
4.1.2.13.1	System shall be able to support multipe analysis type, including numeric, text, characters, calculated, file, Boolean and optional.	Standard
4.1.2.14.1	System shall be able to entry result data manually.	Standard
4.1.2.15.1	System shall support attachment format: Xps, txt, resx, repx, xml, rtf, jpg, jpeg, img, svg, doc, pdf, ico, png, lbl, resources, caf, csv, log, msg, smpkg.	Standard
4.1.2.21.1	System shall be able to record multiple users against a test or result.	Standard

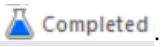
3.2.1.6.3 Function Design

Requirements Description :

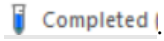
Users can record results entry for single or multiple tests performed on multiple samples. Users can record multiple result type and can upload attachments. And system can unlimited user to enter irregularities or partial or incomplete results and mark result against MLP.

Design Overviews :

a. **Folders and mater menu**
   Add Result entry folder.
   Add Result review folder.
b. **Functional logic**
   While all the results of the sample is saved completely:

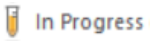
   i: the samples status automatically update as " completed" 

.

   ii: the tests status automatically update as " completed" 

.

   While partial results of the test is saved:

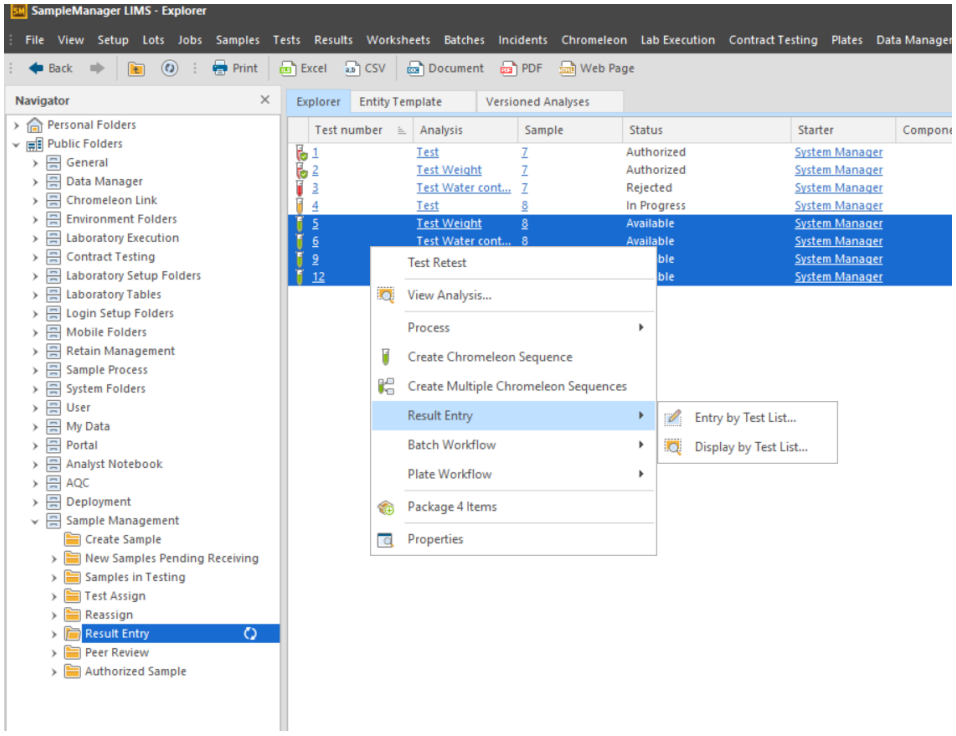
   i: the test status automatically update as " In Progress" 

Commented [NR71]: In case there is a preparation phase after the receipt of physical sample, business wants to show it as "In progress"

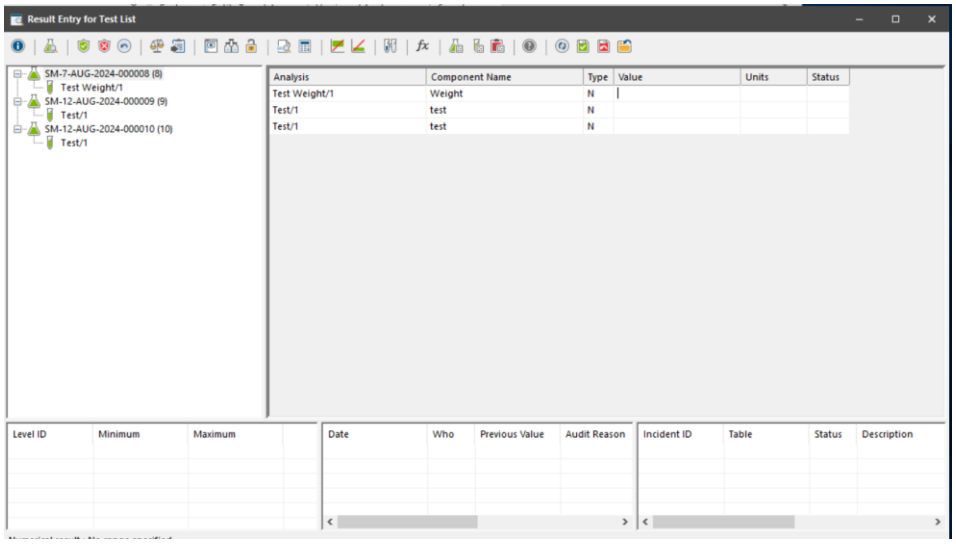
Commented [CG72R71]: Discussed before

Function Entrance :

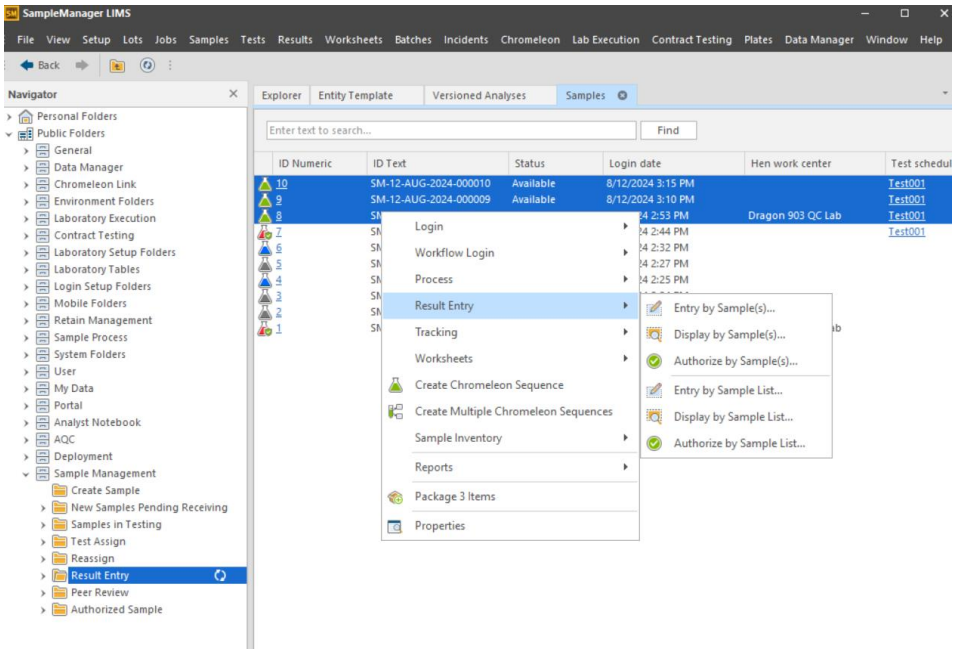
Sample Management Cabinet → Result entry Folder→Select single or multiple tests and right click menu "Result Entry" →Select menu "Entry by test list" as shown here.



And then will pop up result entry window as shown below.



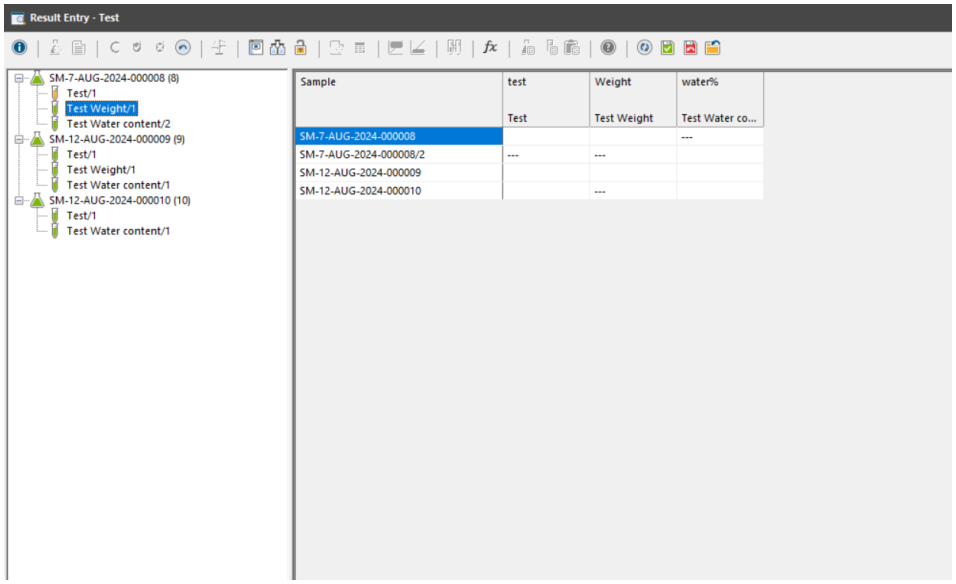
Under navigation bar→ select sample→ Select "Result Entry" →Select menu "Open Samples" as shown here. Select single or multiple samples RMB "Entry by sample(s)".



And then will open result entry window as shown here.

Commented [NR73]: what is RMB?

Commented [DL74R73]: RMB is right mouth button



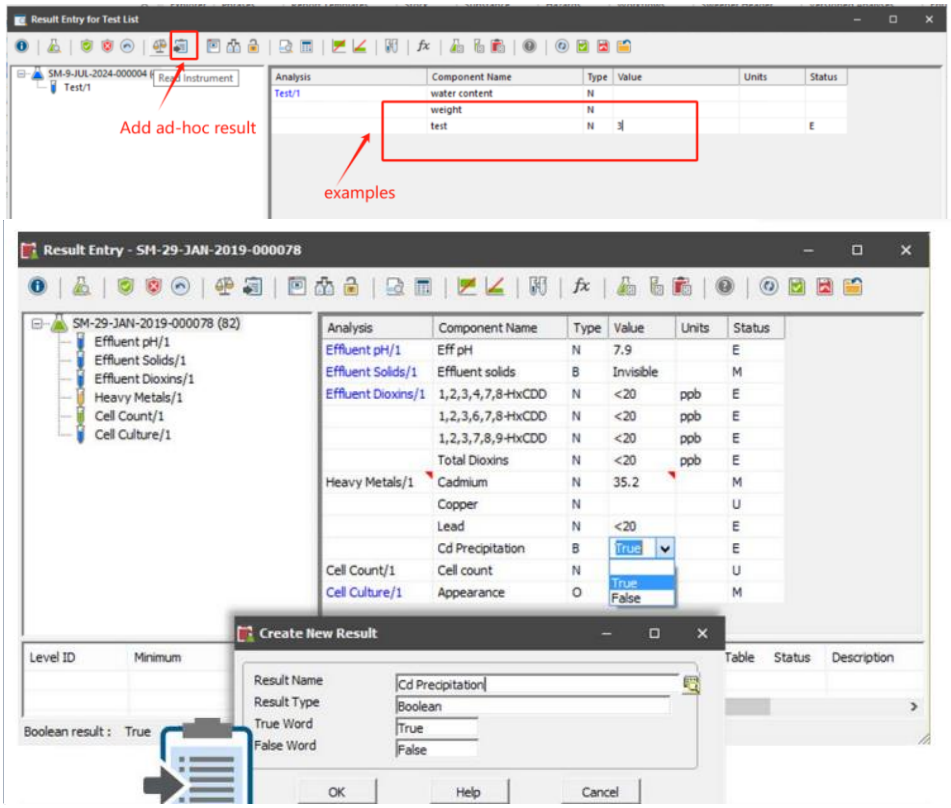
Interaction Events:

- Four workflows in Phase A are all adapt to this operation, therefore will not description in detail here.***Result entry two ways***

  By Sample: the result entry screen is tabular and results are entered into cells by analysis and component.

  By Test: the result entry screen is tabular and results are entered into cells by test and component.

- ***The Results Entry screen has two formats***
  Depending on the menu option chosen. The operations of both formats are the same. Results are entered into the value / component column as shown. The Status column displays the result status after entry of a result.
- ***Modify results***
  Results can be modified at this screen. The user will be asked to confirm any modifications. It is not necessary to enter all of the results in one session. Results can be entered and modified at any time (E or M). To save results to the database click the save button.
  Copy results
- ***Add ad-hoc results***
  Not all tests are routine, for Non-Standard testing, user usually needs to add some temporary tests, so user can use add hoc results function to meet their requirements.



Sometimes you will need to add a result to a sample that is not defined in the analysis that creates the test. This is called an Ad-Hoc result. Consider a sample that arrives in the laboratory, and it does not conform to the normal situation. For example, the sample appears to have solid particles suspended in it and this is not usual as shown above. Enter (create) a result name and appropriate values into each field based on the result type selected.

3.2.1.7 Copy Result

3.2.1.7.1 Business Scenarios

For some OQC sample, based on requirements, system can support copy IPQC last phase sample results to OQC sample results, and then the OQC sample will not need to fill in results anymore manually.

3.2.1.7.2 RTM Mapping

NA

Commented [FS75]: What is true word and false word in the screen shot

Commented [MX76R75]: Just a phase optional for results, can configurable

Commented [DL77R75]: Is this means "true word" & "false word" is actually meaningless and can be replaced by other field?

Commented [DL78]: Copy IPQC to OQC available for all phases or it will only available to the last phase. If it's only available with last phase how do system know which phase is last phase?

Commented [ZZ79R78]: The sample Authorized status is the final stage of the sample, and if copy result is used for all stages, it makes no sense to copy the sample when there are no results or only partial results.

Commented [DL80R78]: Agreed, but how can QC LaD differentiate which phase is the last phase?

Commented [WX81R78]: as discussed ealier with Neelima, there are two options:
1: SAP end directly provide the indicator of last phases within each IPQC inspection lot.
2: if above not viable, when click on the "copy to OQC " menu, LIMS will check all the IPQC sample who share the same inspection lot no. and with all the samples (corresponding to each phase number of the same Inspection lot), LIMS to further compare their phase no, and see if current sample has the biggest phase no.

Commented [ZZ82R78]: Options 1

Commented [NR83R78]: There was agreement to provide whether activity is IPQC or OQC. We won't be able to provide which is the first and last. Normally it is the order of ascending.

Commented [ZZ84R78]: The current solution is that when SAP sends out the Inspection lot, it provides an indication of whether or not copy results are allowed, and QCLAD uses this indication to determine whether or not the results can be copied to the OQC samples.

Commented [NR85R78]: Unfortunately this may not be possible as the master data may not be possible. Can the decision be made by the user whether it is last phase or not.

Commented [WX86R78]: 1: **to control the visibility of "Copy result" menu?**
for the time being, This menu should be visible to all the IPQC sample.
later on, if any further restriction is needed, can treat this as enhancement.

2: **after click on the menu, to further determine if the last phase IPQC sample**
if above not viable, when click on the "copy to OQC " menu, LIMS will check all the IPQC sample who share the same inspection lot no. and with all the samples (corresponding ...

3.2.1.7.3 Function Design

Requirements Description :

The user manually selects the IPQC samples that need to be copied to the OQC, and the system will match the OQC samples with the same lot number as the SAP inspection lot for that IPQC and copy the results to the OQC samples and determine the results based on the specification;

Design Overviews :

a. **Folders and mater menu**
Add Sample Authorized folder.

d. Table change
Add new fields in Sample table to "phrase no" (text field) from SAP Inspection lot.
Add new fields in Sample table to "result from" (links to sample)

Function Entrance :

Sample Management Cabinet → Sample Authorized Folder→Select single sample and right click menu "Copy Result To OQC".

1. Need to add a master menu "Copy Result", which should be only visible to all IPQC sample.

Note: the time being, this menu should be visible to all the IPQC sample. Later on, if any further restriction is needed, can treat this as enhancement.

2. After click on the "Copy Result" menu, system should query all the IPQC sample, regardless what status it is, whose inspection lot no is the sample with current sample. And then to further check the among all the queried samples if the "phase no" of current sample is the biggest one, if biggest one then seen as last phase, then system can proceed the next logic. If not, will prompt warning message and terminate the copy actions.
3. The system will look for samples with an Inspection Type of OQC (Phrase no :0010) based on the SAP Inspection Lot number of the selected sample, and give a reminder if it does not exist or if more than one exists, and will not copy the results;
4. The system compares the test items in the selected samples with the test items of the OQC samples found, and gives a reminder if there is a discrepancy, and does not copy the results;
5. Instead, the full results (excluding the added Ad-Hoc results) are copied to the OQC sample and determine the results based on the specification;

After the results are copied into the OQC sample, system also needs to copy the IPQC sample ID into the field "Result From" of the target OQC sample.

Commented [FS87]: Is the purpose of adding this folder that is convinent for QC to filter all the IPQC samples that can copy result to OQC?

Commented [ZZ88R87]: Yes

Commented [DL89]: User shall be able to select the right sample to be copied to OQC

Commented [ZZ90R89]: The allowed copy flag comes from SAP's Inspection lot and is determined by this identification field

Commented [NR91R89]: Allowed copy flag will not come from SAP inspection lot. OQC, IPQC information will come from inspection plan master data from SAP

Commented [DL92]: In inspection plan in SAP, if there is only OQC(act #: 0010) then the copy result function shall not be seen. While if IPQC(other act. #) is under inspection plan level the copy result shall be available.

Commented [ZZ93R92]: The allowed copy flag comes from SAP's Inspection lot and is determined by this identification field

3.2.1.8 Test Retest/Resample

3.2.1.8.1 Business Scenarios

User need to retest a test item if there is any mistake or strange data. System need to create retest without lossing original test record.

User need to trigger resample if they need a new sample. Workshop will send a new sample and then lab will start to work on the new sample.

3.2.1.8.2 RTM Mapping

URS Ref.	Requirement	Type
4.1.2.22.1	System shall be able to retest without loss of original data.	Configuration
4.1.2.23.1	System shall be able to enforce entry of a justification if there is a retest/resample.	Configuration

3.2.1.8.3 Function Design

Requirements Description :

System allow user to retest a test item or trigger resample in QCLad.

If a test is triggered retest, a new same test will be created and itself will be rejected automatically.

If a sample is triggered resample , system will create a new sample and copy sample properties.

When retest or resample is triggered, system will ask a mandatory comment.

Design Overviews :

a. **Table changnchange**
   Add new fields in test table, to record retest date.
   Add new fields in sample table, to record resample date and comment.
b. **Workflow**
   Add nodes in test lifecycle:
   1. Test Retest: create a new same test and reject the origin test automatically
   Add nodes in sample lifecycle:
   2. Resample: create a new same sample. system will create a new sample and copy sample properties including IDH, test items, batch no to new sample. MES ID will automatically updated in new sample. After resample, user can select cancel original samples.
      Detail description as below:

   3. **Resample triggering process:**

      Case 1: before sample review (during inspection, or during sample review)-->**just trigger resample to create another sample but not cancel original sample**-- then

Commented [C94]: 功能待确认

Commented [NR95R94]: Cancelled scenarios should also be added

need to process these two samples respectively on their own-- the result of both sample should be pushed to SAP.

Case 2: before sample review (during inspection, or during sample review)-->**just trigger resample to create another sample but CANCEL original sample manually done in LIMS by user**-- then need to process tnew sample—then push to sAP after reveiwe.

Case 3: after sample review, won't allow resample action.

Case 4: sample amount is not enough for retain after sampel review. It has nothing to do with resmaple. Resample is the concept to ask for additional sample for further additional testing but not for retain.

In terms of the process to create the retain sample, whether munaylly or auto, will be discussed further on PPB.

c. **Folders and Cabinets**
Multiple folders will be added into sample management cabinet for usability:

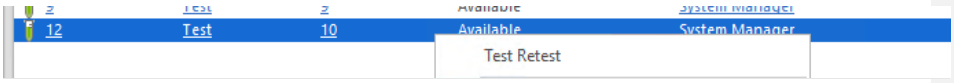
Resampled samples: list of samples that are resampled.

Interaction Events:

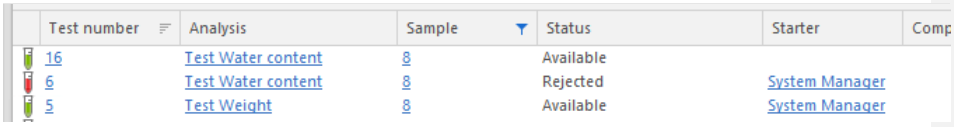
Four workflows in Phase A are all adapt to this operation, therefore will not description in detail here.

- ***Test retest***

User can click 'test retest' RMB in result entry and peer review folders to trigger retest.



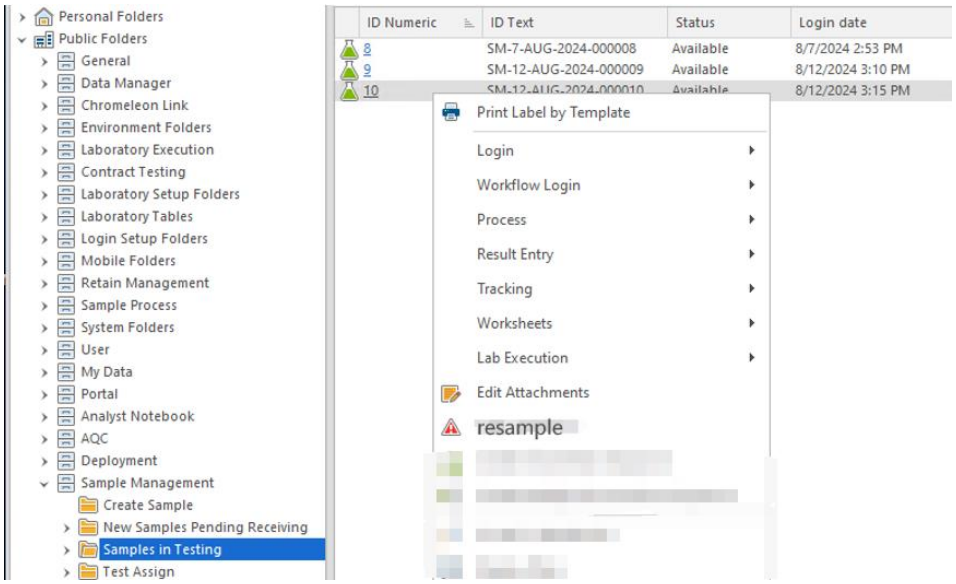
A new test will create and the original one will get rejected.



Test number	Analysis	Sample	Status	Starter	Comp
16	Test Water content	8	Available		
6	Test Water content	8	Rejected	System Manager	
5	Test Weight	8	Available	System Manager	

- ***Resample***

User can click 'Resample' RMB in 'samples in testing' folder. A new sample will be created and it will be in pending receive folder waiting for receiving. The original sample will be stay in the batch. Origin sample will be marked as Is_resampled.



3.2.1.9 *Result Review*

3.2.1.9.1 *Business Scenarios*

While user complete the results entry, usually in order to release the samples need peer review to authorize results here.

Commented [FS96]: Peer review or review can be configurable at plant level.

Commented [MX97R96]: TF will disscuss fesaible way late and inform henkel team via email

3.2.1.9.2 *RTM Mapping*

URS Ref.	Requirement	Type
4.1.2.25.1	System shall be able to enforce peer review by a different user at the material group level.	Configuration
4.1.2.25.2	System shall be able to support peer review by configurable option.	Configuration
4.1.2.26.1	System shall allow the peer review of results by Single sample, Multi test and Multi sample, single test where required.	Standard
4.1.2.27.1	System shall prevent user from further modification on test data after peer review except authorized user and audit trail will be record.	Configuration
4.1.3.1.1	System shall be able to approve individual results.	Standard
4.1.3.1.2	System shall be able to support configure approve option within workflow.	Configuration

4.1.3.2.1	System shall be able to approve single or multiple samples.	Standard
4.1.3.3.1	System shall prevent user from further modification on test data after peer review.	Configuration
4.1.3.3.2	System shall be able to return test to tester if result need modification .	Configuration
4.1.3.4.1	System shall be able to configure system approval functionality at test level and sample level.	Configuration
4.1.3.5.1	System shall be able to approval manually at test level and sample level.	Configuration
4.1.3.6.1	System shall prevent the sample authorization with incomplete testing and alert defined users to the situation in folder.	Configuration
4.1.3.9.1	System shall provide a way to check and indicate the results/samples waiting for approval.	Configuration

3.2.1.9.3 ***Function Design***

Requirements Description :

Users can review and approve results for tests and samples.

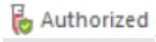
Commented [FS98]: Review and approve tests and samples.

Commented [MX99R98]: confirmed

Design Overviews :

a. **Folders**
   Add Authorized folder.
b. **Functional logic**
   While all the results of the sample is Authorized completely:

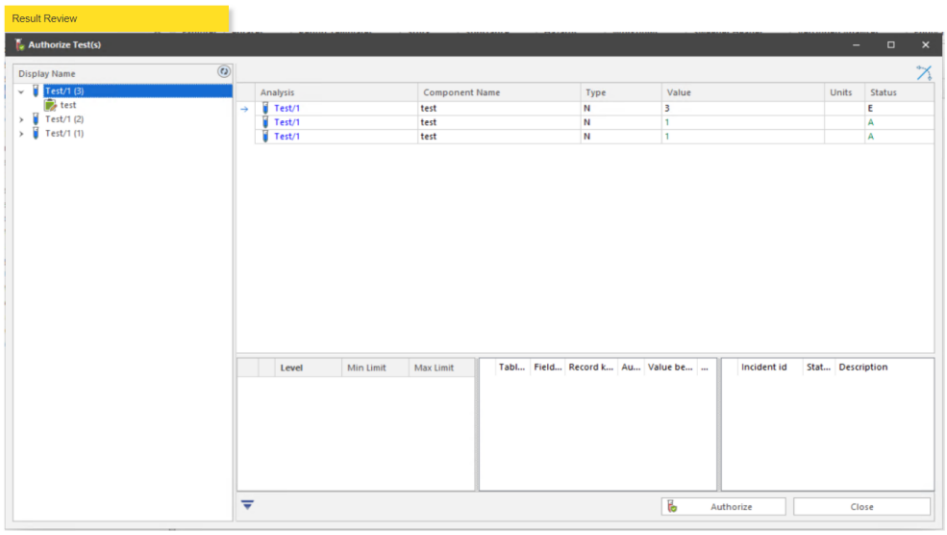
   i: the samples status automatically update as "Authorized"

   ii: the tests status automatically update as "Authorized" 

.

Function Entrance :

Sample Management Cabinet → Result review Folder→Select single or multiple tests and right click menu "process" →Select menu "Authorize" , then will open the authorize window as shown here.



Interaction Events:

Four workflows in Phase A are all adapt to this operation, therefore will not description in detail here.

- ***Result review***

  Click the "Authorize" button to complete tests review operation.

  The system will automatically close the window and tests status will automatically update as "Authorized".

  All the tests of the sample are authorized, the sample will shown in Authorized sample folder.

3.2.1.10 Certificate Generation

3.2.1.10.1 Business Scenarios

After testing result is reviewed and pushed to SAP, generally, testing report or COA will be generated there in SAP using existing report template.

However, it also supports to generate the test report directly in QCLAD when needed.

In a test report, it allows to include the sample information and testing results as designed in the report template; it also allows to include the results from multiple tests and across samples.

3.2.1.10.2 RTM Mapping

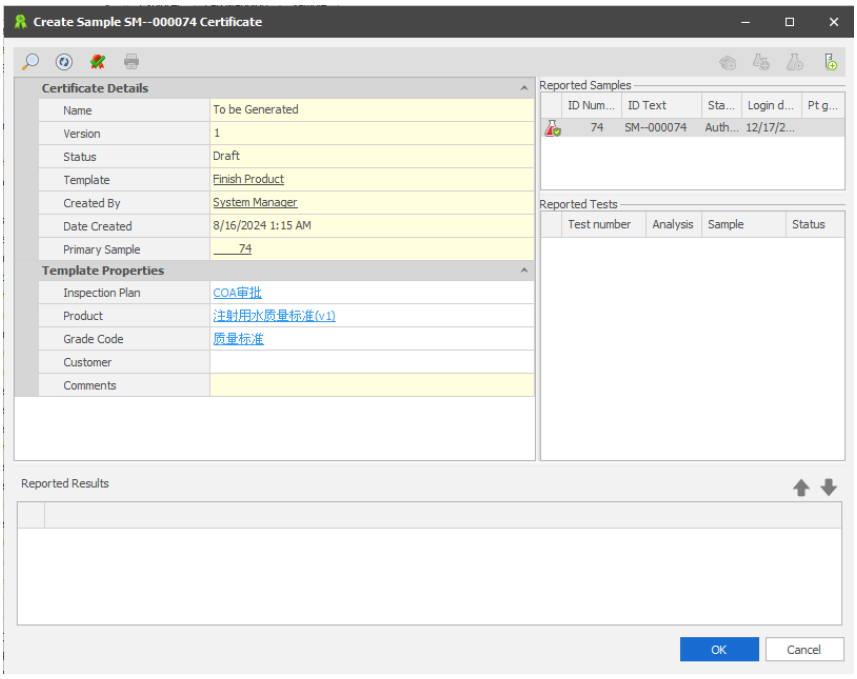
URS Ref.	Requirement	Type
4.1.4.10.1	System shall support to generate single sample report, multiple sample report, COA report.	Configuration
4.1.4.3.1	System must be able to report on the analyzed data at all levels (e.g., raw, recorded and reported analytical data).	Configuration
4.1.4.4.1	System shall collate results from multiple tests for each sample.	Configuration
4.1.4.8.1	System shall be able to generate analytical reports based on the data recorded in QCLAD like date time, QC operators details along with inspection.	Configuration

3.2.1.10.3 Function Design

IQC-Standard RM

After sample is reviewed and authorized, the sample status will be updated to Authorized, and will move to folder “Authorized Sample” folder in QCLAD.

User can right click on any samples from this folder and select “Create Certificate”, and needs to further select a certificate template to proceed the creation.



Note: in QCLAD a testing report or COA is referred as Certificate. And a certificate template is the master data which needs to be maintained and used before certificate creation.

in the creation form, user can

On the template properties section: sample information can be auto fetched and populated here as configured; and user can add in more information as needed. And all these information will be saved as attributes of current certificate and displayed in the certificate as designed accordingly.

On the reported sample/test section: user can add more sample/test into current certificate if allowed in certificate template configuration.

On Reported results section: once range of samples and tests are defined, the results will be displayed here along with their analysis name, result name, value, units, limits.

If everything is good, user can preview the certificate, then click on Apply/OK to create the certificate.

Commented [FS100]: Maybe a diagram is better for understanding

Commented [MX101R100]: confirmed

IPQC without MES

Can refer to point 1.

OQC without MES

Can refer to point 1.

Non-routine sample

Can refer to point 1.

3.2.1.11 Certificate Approval

Commented [FS102]: Don't understand on this part.

3.2.1.11.1 Business Scenarios

After certificate is created, for different types of testing report, such as IPQC testing report, it requires not further approval; yet for OQC testing report, it needs review before approval.

3.2.1.11.2 RTM Mapping

URS Ref.	Requirement	Type
4.1.4.11.1	System shall be able to route the output of reports to screen, printer, file (CSV, XML, PDF) or email.	Standard

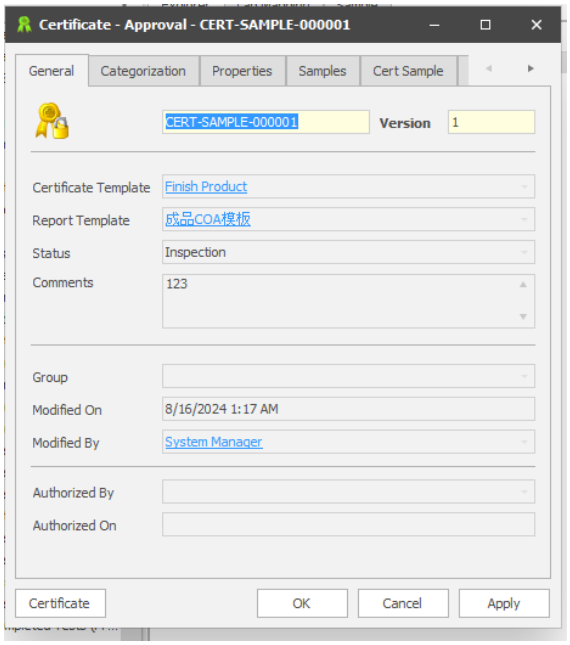
3.2.1.11.3 Function Design

IQC-Standard RM

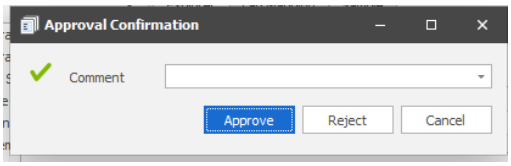
After certificate is created, according to the inspection plan (approval plan) configuration in certificate template, the certificate will be auto approved since no review is required. The certificate will be under “Authorized” status and parked under “Authorized Certificate” folder

Commented [MX103]: Only for non-standard testing need to approval flow, others don't need this

Commented [MX104R103]: Configurable based on sample type



Click “Apply” will pop up window as shown below.



Right click on any certificate, user can view the display in pdf format. And user can further print out the pdf format certificate.

IPQC without MES

Refer to point 1.

OQC without MES

After certificate is created, according to the inspection plan (approval plan) configuration in certificate template, unlike IQC testing report, the certificate will be under "Inspection" status, meaning waiting to be reviewed by the person or role defined in the inspection plan.

And the subjected reviewer can right click on certificate and access to all the information related to this certificate. After review, user can click on OK and proceed approval decision and comment.

Non-Routine sample

Can refer to point 1.

3.2.2 Shift Management

3.2.2.1 Shift Management

3.2.2.1.1 Business scenarios

For project needs, the system needs to maintain the shift information of the plant personnel to serve the later use of the automatic task assignment function; Only need to maintain the on duty information , the user maintains the personnel shift information for a week or more at a time.

3.2.2.1.2 RTM Mapping

URS Ref.	Requirement	Type
4.2.1.8.1	System shall allow user maintain master data for shift schedule by week. Number of shift and shift hour of each shift shall be configurable in the master data at plant level.	Customisation

3.2.2.1.3 Function Design

After comprehensively analysing the shift schedule requirements of personnel, a new shift schedule module needs to be added to the system to allow users to maintain the shift information of plant personnel in QCLAD on a weekly or longer cycle.

Base Master Data:

1 Users need to maintain plant information, and in plant information, this module mainly maintains shift limits: the maximum number of consecutive working days and the maximum number of working hours in a week can be maintained.
Plant Table:

Commented [XW105]: Pending for Henkel to decide whether to include this control/function.

Commented [NR106R105]: We have provided the final details. Kindly update the master data with our requirements

Field	Type	Description
Plant Code	text	should be unique
Description	text	Plant Description
Max working days	integer	the maximum number of consecutive working days
Max working hours	integer	the maximum number of working hours in a week can be maintained

2 Users can maintain the shift type, the shift type table contains: Identity (should be unique, suggest to maintain manually as PlantID + "-" + Code), Code (such as M8, M12, D8), plant, Name (long day shift, long night shift, etc., the user to customize), start time (e.g. 8:30), end time (e.g. 20:30), RemoveFlag (whether to delete the identification);

Shift Type Table:

Field	Type	Description
Identity	text	should be unique, the user to customize, suggest to maintain manually as PlantID + "-" + Code.
Code	text	shift type code,such as M8, M12, D8.
Plant	text	Link Plant Table.
Name	text	shift type name, the user to customize, e.g. long day shift, long night shift, etc.
start time	interval	Working hours for shift types, e.g. 8:30.
end time	interval	Off work hours for shift type, e.g. 20:30.
RemoveFlag	boolean	whether to delete the identification.

2.1 The system supports for shift type data maintenance, and it can be added, removed, modified, displayed;

Main Master Data:

1 The system should support that users can create a shift schedule for a group for one week (or two weeks, with a custom period) that contains: Id, Name, Description, Shift Period (start date, end date), Modifier, Modified Time, Plant, Enabled (indicates whether the scheduling is confirmed to be in effect), RemoveFlag (whether to delete the identification).

shift schedule table :

Field	Type	Description
Identity	text	should be unique, the user to customize.
Name	text	shift schedule name, the user to customize.
Shift Period Start date	Date	Shift Period Start date, e.g. 2024/08/05.
Shift Period End date	Date	Shift Period End date, e.g. 2024/08/09.
Modifier	text	Modifier for this shift schedule, Link Personnel Identity.
Modified Time	Date	Modified time for this shift schedule.
Plant	text	Link Plant Table.
Enabled	boolean	indicates whether the scheduling is confirmed to be in effect.
RemoveFlag	boolean	whether to delete the identification.

2 A shift schedule can contain shift information for multiple people within the scheduling cycle, the same day can be maintained for more than one person more shift type of shift information.

2.1 The shift information contains: order number, personnel (Links Personnel Identity), shift type, shift start time (in datetime format), shift start time of the year, month, day, week, shift end time(in datetime format), the end of the shift time of the year, month, day, week, shift duration (unit: hour), description.
shift information table :

Field	Type	Description
Identity	text	shift schedule, Links shift schedule Identity.
order number	text	should be unique, System automatically generated.
personnel	text	shift worker, Links Personnel Identity
shift type	text	Links Shift Type Identity
shift start time	Date	start of work, e.g. 2024/08/05 8:30
shift start time of the year	integer	Year of the working hours, e.g. 2024
shift start time of month	integer	Month of the working hours, e.g. 8

Commented [XW107]: Henkel: should be only showing Identity of those operators in shift management.

Should not display the name here up front here.

Commented [MX108R107]: Confirmed by both paties

shift start time of day	integer	Day of the working hours, e.g. 5
shift start time of week	integer	Week of the working hours, e.g. 1
shift end time	Date	end of work, e.g. 2024/08/05 20:30
shift end time of the year	integer	Year of the off work hours, e.g. 2024
shift end time of month	integer	Month of the off work hours, e.g. 8
shift end time of day	integer	Day of the off work hours, e.g. 5
shift end time of week	integer	Week of the off work hours, e.g. 1
planned hours	integer	planned hours (unit: hour), The hour difference between start of work and end of work.
actual hours	integer	The first time the scheduling information is created it is consistent with the planned hours field, which is automatically calculated by the system
description	text	shift information Description.

Commented [NR109]: Please consider the master data provided by Henkel

Commented [XW110R109]: The shift start time or end time should be interacted by user.

The remaining red highlight ones is auto calculated by system itself for future calculations. Require no end user to key in.

Commented [MX111R109]: Confirmed by both partied

2.2 Because a scheduling can maintain shift information for a period of time such as one week or two weeks, the system supports displaying a calendar-formatted interface for the user to maintain shift information for each day of the cycle when maintaining shift information in the scheduling message;
For each day in the calendar, it displays all the shift information of the day, the shift start time (displaying the time is sufficient, not the date), the shift end time (displaying the time is sufficient, not the date), the ID of the personnel, and the shift type Code.

2.3 When you select a day in the calendar by right clicking on it, you need to have the ability to add, modify, and delete in the RMB
When a user maintains shift information and selects a date in the calendar to right-click to add or modify the shift information, after selecting the shift type, the start time and end time of the shift type will be automatically brought into the start time and end time of the shift, and the date of the start time and end time of the shift will be the current date selected in the calendar, and changes are allowed;

Commented [NR112]: Display of user should be based on user id, no personal name nor email address should be displayed.

Commented [NR113R112]: Only need to maintain the availability of operators in shift, no need to include vacation and leave information as this is all HR work.

Commented [MX114R112]: Confirmed by TF

2.4 When the user maintains the shift information and clicks Save, the year, month, day and week fields of the shift start time will be written into the corresponding fields respectively; the year, month, day and week fields of the shift end time will be written

into the corresponding fields respectively, and the user is not allowed to change them manually, and there is no need to display them on the interface; this is to facilitate the Excel export of the shift information later.

2.5 When a user maintains shift information and clicks Save, the planned hours and actual hours (in hours) is automatically calculated;
The planned hours calculation rule is: the number of hours obtained from the end time of the shift - the start time of the shift;
The actual hours calculation rule is: Consistent with the rules for calculating planned hours;

2.6 When the user maintains shift information and clicks Save, the system needs to verify whether the same person has more than one shift information on the same day, and if so, determine whether there is any time overlap in the information of more than one shift, and if there is any time overlap, then it will give a prompt and will not be submitted for saving;
Time overlap judgment logic: for example, Zhang San in 2024-7-19 8:30 to 2024-7-19 20:30 shift information type on shift, and Zhang San in 2024-7-19 13:30 to 2024-7-19 20:30 shift information type in the vacation, in the 2024-7-19 13:30 to 2024-7-19 20:30 the same If there are two trips in the same time period from 2024-7-19 13:30 to 2024-7-19 20:30, they are counted as overlapping;

2.7 When maintaining personnel scheduling, the number of working hours per natural week (Monday to Sunday) is restricted to be less than or equal to the maximum maximum number of working hours in a week in the plant, and the maximum number of consecutive working days is less than the maximum number of consecutive working days in the plant, and if the restriction is exceeded, a prompt will be given, and it will not be submitted for saving.
Calculation logic for continuous working days: query the number of entries of the shift information of the current personnel from the current day to the previous day, which are of the type of on-shift;

Calculation logic of working hours in a week: query the current personnel in the natural week of the day (for example, the day is Wednesday, then query the time from Monday to Wednesday) shift information table in the type of shift information in the shift in the class of the shift length of the cumulative addition;

Commented [XW115]: Pending for Henkel to decide whether to include this control/function.

Commented [NR116R115]: Confirmed from Henkel only consecutive working hours/week should only be considered.

Commented [MX117R115]: TF confirmed.

2.8 The system supports adding, deleting, modifying and querying the Shift Scheduling information maintained by the user.

2.9 To create a new shift plan (essentially, is to create to planning for future time)
a) Each shift will auto calculate the value of "current week", "planned hour", "actual hour" (by default, actual hour is the same as planned hour)based on the "Operator+ Day+ Shift Type"； be noted that each shift is breakdown by "Operator + Day".

b) When apply to save, system will validate that, for each operator, if the total Actual hours of current week exceeded the maximum number of working hours in a week can be maintained (the variable configured on each plant). if yes, prompt warning and not allow saving.

c) When apply to save, system will validate that, for each operator, if the consecutive work day of current week exceeded the maximum number of consecutive working days (the variable configured on each plant). if yes, prompt warning and not allow saving.

2.10 To update a created shift plan

a) To add new shift for new operator or existing operator, will follow all the logic on case 2.9.

b) To update the existing shift, one is allowed to modify the "Actual Hour" and key in "Comment".

c) If the existing shift being updated is on current day or before current day, system will validate that, for each operator, if the total Actual hours of current week exceeded the 60 hours (the variable configured on each plant). if yes, prompt warning yet allow saving , given system should allow record what has happened.

d) If the existing shift being updated is after current day, system will validate that, for each operator, if the total Actual hours of current week exceeded the maximum number of working hours in a week can be maintained (the variable configured on each plant). if yes, prompt warning and not allow saving.

2.11 To delete a created shift plan

Maintenance personnel with deletion permissions can delete the already created shift information, the system does not impose restrictions.

3 The system should support the export of shift schedule data to Excel:

The user selects the time range and exports the personnel shift information: personnel ID, personnel Name, personnel work centre , shift type ID, date (date as table header), Displays the shift type and actual hours of the shift information.

Examples are shown below (ignore the background colour):

The screenshots are for reference only, the final design may differ from the screenshots in the document.

Commented [NR118]: We can skip the department as shown in the below image, but it should be based on the work center. The shift supervisor role should be performing shift management based on work center level.

Commented [MX119R118]: Group by workcenter; group is standard function; 1.department column needs to change as workcenter;
2.Use ID instead of GPN, and column name only userid should be employee id.

Print out

GPN 工号	Name 姓名	Department 部门	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
			Wed	Thu	Fri	Sat	Sun	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Mon	Tue	Wed	Thu	Fri
	zhangsan	hr	A8	A8		D8	M12			D8	D8	D8	D8	D8			D8		D8	D8	M12			D8	M12	D8	D8	D8			D8	D8	D8
			D8	D8	D8		D8	M12	N12				M12	M12			D8	M12	M12			M12	M12	M12			D8	M12	N12			D8	
			M12	N12	N12			M12	M12	N12				M12	N12			D8	M12	N12			M12	M12	N12			M12	M12	N12				
				M12	M12	N12			M12	M12	N12				M12	N12			D8	M12	N12			D8	M12	N12			M12	M12	N12			
				M12	M12	M12	N12				M12	N12				M12	N12			D8	M12	N12			D8	M12	N12			M12	M12	N12	
			N12			D8	M12	N12	N12			M12	N12				M12	N12			D8	M12	N12			D8	M12	N12			D8	M12	

3.2.3 Master Data

3.2.3.1 Production Line

3.2.3.1.1 Business Scenarios

Currently in each Henkel's plants, the semi-finished product and finished product are classified with a production line, for example MPT,TGS and PAS etc. however the raw material or packaging material, which are purchased from suppliers, is not yet categorized the same with production line.

And production line is not yet standardized across all plants with Henkel, the same products can be classified into different production lines in terms of naming even though their meanings are the same essentially.

For PPA, production line will be maintained by each plant respectively. In the future, if needed, Henkel will introduce another concept of Technology to standardize the naming of all the production lines within all the Henkel' s plants.

Furthermore, production line information is not maintained in material master data in SAP; is maintained in MES only.

3.2.3.1.2 RTM Mapping

URS Ref.	Requirement	Type
N/A	N/A	N/A

3.2.3.1.3 Function Design

3.2.3.1.3.1 Production Line Maintenance

1. User should be able to manually maintain production line master data in QCLAD. And for each production line, user can maintain below information:

Table Field Name	Comment

Production Line	The name of the production line, **part of the unique key**.
Plant	The plant in which the production line is used, **part of the unique key**.
Description	Plain text field

2. User should be able to modify, display, and remove a work centre master data if given the privilege.

Commented [NR120]: Please change the workcenter to production line

3.2.3.1.3.2 Relation between Production line and Material

For all the material master data synced from SAP down to QCLAD, on plant level, user should manually assign a production line to the material master.

- If semi-FP or FP material, just assign the adequate production line by selecting.
- If raw or packaging material, just select the generic production line given currently these type material is not classified with any production line.

3.2.3.1.3.3 Relation between Production Line and Sample

When an inspection lot is pushed from SAP to QCLAD, normally one sample will be created.

- if the material of that inspection lot is already linked with production line in QCLAD, the production line will be auto fetched and populated into the sample created for the inspection lot.
- If the material of that inspection lot is not yet linked with any production line in QCLAD, user can modify the sample and manually select the production line.

3.2.4 Dashboard

3.2.4.1 Static Dashboard

3.2.4.1.1 Business Scenarios

The system can query sample information based on Plant, work center, business team (group) filter conditions, and statistical display of the number of samples of each state, and the number of samples of each product line and type of test and other statistical displays;

3.2.4.1.2 RTM Mapping

URS Ref.	Requirement	Type
4.11.1.1	Two dashboards are required, a static dashboard with filtered conditions and a dynamic dashboard with default conditions with timed refreshes; the content displayed in both dashboards is the same	Customisation
4.11.1.2	The static dashboard with filters has the following filters:plant, work center, business team(group)	Customisation
4.11.1.3	Dynamic dashboards with default conditions are filtered by plant, work centre, business team(group) where the currently logged in person is located.	Customisation

URS Ref.	Requirement	Type
4.11.1.4	The dashboard shall display: Date of the day, of that day, the lists of on-duty operators, only showing the operator ID	Customisation
4.11.1.5	The dashboard shall display: of the criteria, the statistic of all the tasks : Count of samples received on the same day that were not reviewed and completed Count of samples received prior to the day that were not audited and completed Count of samples completed Count of samples in-testing Count of not started Count of delayed sample Count of urgent sample	Customisation
4.11.1.6	The dashboard shall display: list of delayed sample	Customisation
4.11.1.7	The dashboard shall display: list of urgent sample	Customisation
4.11.1.8	The dashboard shall display:the column chart in which x axis has the Sample Count, and y axis has the product line. and the column chart is coloured according to the status of the samples (Not Started, Test in Progress, Test Completed).	Customisation
4.11.1.9	The dashboard shall display: anotther column chart in which x axis has the Sample Count, and y axis has the inspection type (IQC, OQC, IPQC), and the column chart should be colour-coded according to the sample status (Not Started, Test in Progress, Test Completed).	Customisation

3.2.4.1.3 Function Design

Dashboard interface reference (screenshot for reference only):

Commented [NR121]: Legends should be displayed in bar graph
Below table of sample need to include material, batch number. Will team be production line? If so, should text be changed to production line? Is it possible to select older dates? Can we change the button from “Search” to “Apply filters”. Whenever user wants to get updated data, should we click on “Search”?

1、 Clicking the "Apply filters" button on the dashboard allows you to query sample information (excluding authorization status) based on three filter conditions: plant, work center, and business team (group), as well as personnel information under day shift conditions.

2、 Dashboard interface shows the "Current Date".

3、 Based on the query conditions for personnel information searched by the day shift, the interface will display only the personnel ID..

4、 Based on the Non-Authorized samples identified by the specified query conditions, the following information has been compiled.:

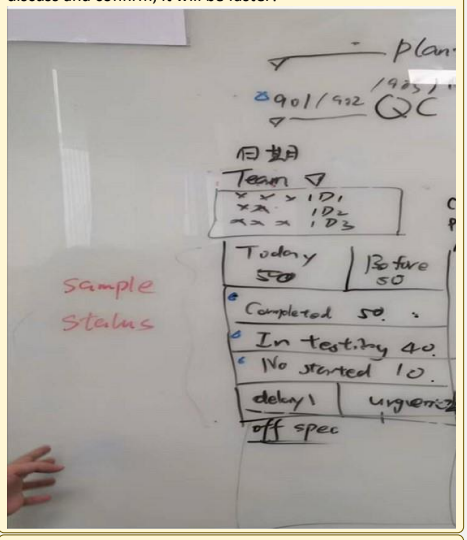
Commented [ZZ122R121]: The legend can be displayed, as I have added to the detailed description later;
The sample table can show more fields, which I have added to the detailed description later;
A team is not a production line, it's a group.
According to the current requirement, it is not possible to select the date, because the statistics in the interface are all the current unfinished samples;
The text of the search button 'Search' can be displayed as 'Apply filters'; when the user changes the filter information, clicking on this button will refresh the data display in the interface;

Commented [DL123R121]: Hi Zhenjuan, we are expecting to group via production line, could you please give some example for the "group" you mentioned?

Commented [DL124R121]: Inside the barcode, the number shall be shown

Commented [DL125R121]: Please be detailing with wording. E.g. Before shall be backlog from previous day

Commented [ZZ126R121]: The group is the data segregation in our system, it should correspond to your business teams; this is also according to what we discussed before, if there are still questions we can make a meeting to discuss and confirm, it will be faster!



Commented [ZZ127R121]: Before is not the previous day's sample volume, but the volume of samples received before that day that have not been audited.

Commented [DL128]: Please elaborate this item

Today	Before
50	30
Completed	50
in–testing	40
not started	10
delayed	urgent
10	10

a) The count of samples received on the same day but not yet authorized, the title is today;
b) The count of samples that were received prior to the day but were not authorized, the title is before;
c) The count of samples whose status is completed, the title is completed;
d) The count of samples with the status of in testing, the title is in-testing;
e) The count of samples whose status is not started, the title is not started;
f) The count of delayed sample, the title is delayed;
g) The count of urgent sample, the title is urgent;

5、Based on the specified query conditions, the following list of sample information is retrieved from the dataset.:

delayed sample

Sample ID	Sample Name	Status	material	batch number	.....

urgent sample

Sample ID	Sample Name	Status	material	batch number	...

a) The list of delayed sample
b) The list of urgent sample

The sample table can show more fields such as material and batch number etc.

6、The unauthorised samples queried according to the query criteria are counted and displayed as a bar chart as follows:

Commented [HS129]: The sum of Completed, in-testing, not started should be the sum of Today and before.

Commented [DL130]: Is the detailing can ben viewed via rolling tab?

Commented [ZZ131R130]: Yes in Static Dashboard, no in Dynamic Dashboard.

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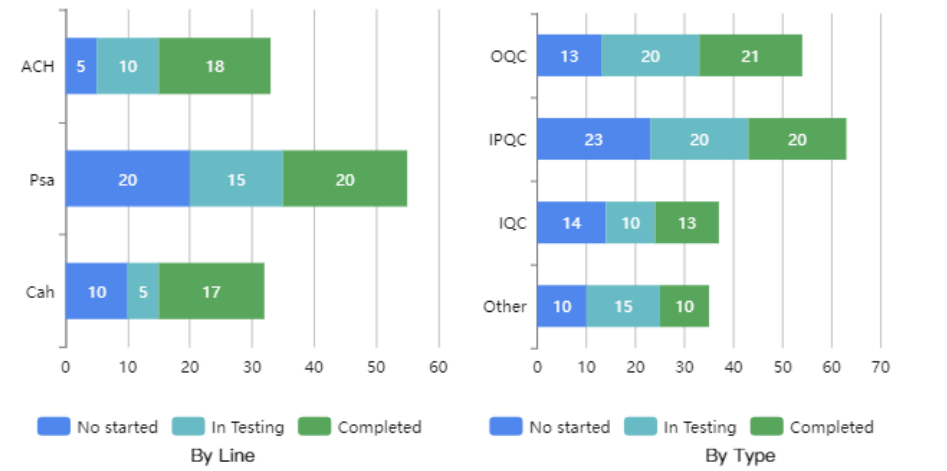

a) The queried samples are grouped according to the product line and displayed in a bar chart according to the status (not started, in testing, completed); the horizontal coordinate is the number of samples, the vertical coordinate is the product line, and the samples are displayed in colors according to their status, and display the legend.

b) The queried samples are grouped according to the test type and displayed in a bar chart according to the status (not started, in testing, completed); the horizontal coordinate is the number of samples, the vertical coordinate is the test type, and the samples are displayed in separate colors according to their status,and display the legend.

3.2.4.2 Dynamic Dashboard

3.2.4.2.1 Business Scenarios

The system automatically queries the sample information according to the Plant, work center, business team (group) where the currently logged-in user is located, and carries out statistics, displaying the number of samples in each state, and the number of samples in each product line and type of test statistics; and carries out automatic refresh at regular intervals;

3.2.4.2.2 RTM Mapping

URS Ref.	Requirement	Type
4.11.1.1	Two dashboards are required, a static dashboard with filtered conditions and a dynamic dashboard with default conditions with timed refreshes; the content displayed in both dashboards is the same	Customisation
4.11.1.2	The static dashboard with filters has the following filters:plant, work center, business team(group)	Customisation

URS Ref.	Requirement	Type
4.11.1.3	Dynamic dashboards with default conditions are filtered by plant, work centre, business team(group) where the currently logged in person is located.	Customisation
4.11.1.4	The dashboard shall display: Date of the day, of that day, the lists of on-duty operators, only showing the operator ID	Customisation
4.11.1.5	The dashboard shall display: of the criteria, the statistic of all the tasks : Count of samples received on the same day that were not reviewed and completed Count of samples received prior to the day that were not audited and completed Count of samples completed Count of samples in-testing Count of not started Count of delayed sample Count of urgent sample	Customisation
4.11.1.6	The dashboard shall display: list of delayed sample	Customisation
4.11.1.7	The dashboard shall display: list of urgent sample	Customisation
4.11.1.8	The dashboard shall display:the column chart in which x axis has the Sample Count, and y axis has the product line. and the column chart is coloured according to the status of the samples (Not Started, Test in Progress, Test Completed).	Customisation
4.11.1.9	The dashboard shall display: anotther column chart in which x axis has the Sample Count, and y axis has the inspection type (IQC, OQC, IPQC), and the column chart should be colour-coded according to the sample status (Not Started, Test in Progress, Test Completed).	Customisation

3.2.4.2.3 Function Design

The interface and display content are almost the same as Static Dashboard, without the search button. The filter conditions of Plant, work centre and business team are automatically brought into the corresponding information display of the currently logged-in personnel, and no change is allowed.

Commented [NR132]: Here how the filters work? If user has selected one and kept will it be auto refreshed?

Commented [ZZ133R132]: The data filter here is automatically brought into the login user's Plant, work centre, team (group) automatic query, the user can not manually modify the filter conditions in this interface.

The system will automatically bring in Plant, work centre, business team to query the unauthorized Sample information, and under the condition of the day duty personnel information; and statistics and display and Static Dashboard consistent.

The Dynamic Dashboard interface needs to be regularly (at 5-minute intervals) refreshed with the queried data, along with the corresponding statistics and displays.

Commented [DL134]: How long is the time interval for regularly?

~~3.2.4~~3.2.5 Resource Management

~~3.2.4.1~~3.2.5.1 Stock Management

~~3.2.4.1.1~~3.2.5.1.1 Business Scenarios

A stock record contains the details of a type of consumable that you want to track.

A stock batch records the details of a specific physical quantity of a stock. Stock batches can be created for:

·Each bottle of a laboratory reagent

·Each manufacturer's batch of a specific diagnostic test kit

~~3.2.4.1.2~~3.2.5.1.2 RTM Mapping

URS Ref.	Requirement	Type

4.1.6.1.2	System shall be able to support QC retain sample, consumables and spare parts can select location.	Standard
4.1.6.1.3	System shall be able to maintain safety stock of consumables and spare parts.	Standard
4.1.6.1.4	System shall be able to maintain storage condition for retain samples, consumables and spare parts.	Standard
4.8.1.6.1	System shall alert specific user when chemicals or instruments is/are expired list in folder.	Configuration
4.8.1.7.1	System shall have capability to maintain master data for consumable, spare parts, equipment.	Standard
4.8.5.1.1	System shall support stock management to maintain information including characteristics (labels, location, available amount), ordering (supplier, expiry warning, minimum quantity (safety stock) alert), inventory, parameters for use in calculations.	Configuration
4.8.5.1.2	System shall support stock management to maintain stock type including retain samples, consumables, standards, reagents, instruments, spare parts.	Configuration
4.8.5.1.4	System shall be able to set and configurate minimum quantity in stock management.	Standard
4.8.5.1.5	System shall be able to set and configurate maximum quantity in stock mangement.	Customisation
4.8.5.1.6	System shall support to print stock label.	Standard
4.8.5.1.7	System shall automatically deduct the stock quantity when consumption.	Configuration
4.8.5.5.1	System shall be able to maintain the expiry date field of stock.	Standard
4.8.5.5.2	System shall be able to display the expiry date field of stock in folder.	Configuration
4.8.5.6.1	System shall be able to be warning for expiry stock in folder.	Standard
4.8.5.6.2	System shall be able to be warning for Low amount storage in folder.	Standard
4.8.5.8.1	System shall support to maintain stock information, including handling and storage precautions, security data, associated hazards and risks and GHS requirement.	Configuration

~~3.2.4.1.3~~3.2.5.1.3 Function Design

Requirements Description :

Users can maintain Stock information.

Users can view near expiry stock and low amount of stock and maximum of stock.

Users can print stock label.

Design Overviews :

a. Table change

Add maximum field under table stock.

b. Folders and mater menu

Add Stock Management Cabinet.
Add StockBatch folder.
Add Reminder of low amount folder.
Add Warning for expiration folder.
Add Reminder of maximum amount folder.

c. Functional logic

Stock Batch Quantity automatically deduct the stock quantity when consumption.

Function Entrance :

[Stock]

Stock Management Cabinet → Stock Folder→ Right click Add button, will pop up Stock-Add window as shown here. Note: Stock is static data, need to maintain stock information here at first.

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Commented [HS135]: Maximum amount (like explosive reagents) should also be considered.

Commented [MX136R135]: It's out of scope according RTM

Commented [NR137R135]: I remember adding that as extra during Gap analysis, can you please confirm.

Commented [MX138R135]: Sorry misunderstanding, we have checked, maximum is in scope, we add it here now

Commented [NR139R135]: Thanks Mengran for confirming. Closing it

Commented [DL140]: Please update the FS with max amount

Commented [MX141R140]: Done

Commented [NR142]: Is this applicable for retain samples as several details may not be available there.

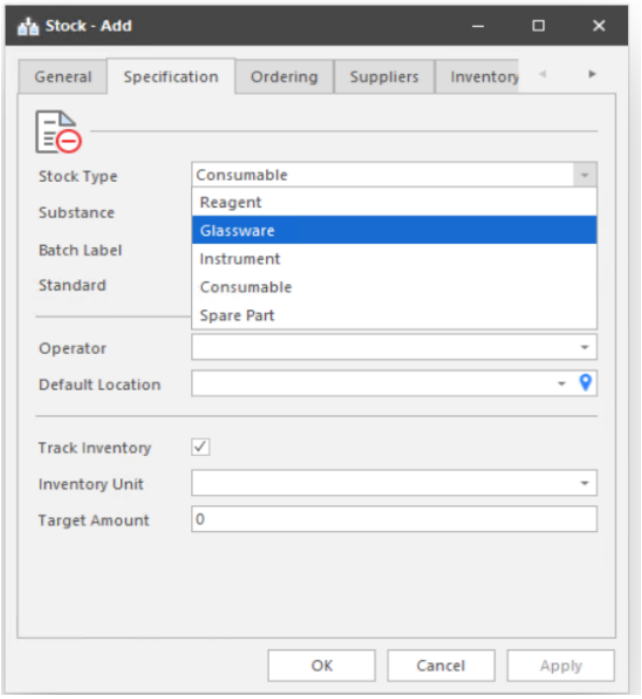
Commented [MX143R142]: Yes, Retain sample is individual module, we can maintain retain sample in retain management.

Commented [NR144R142]: So, this section only for consumables, equipments right?

Commented [MX145R142]: Right!

Commented [NR146]: How to define locations in QCLAD? Is it a part of master data?

Commented [MX147R146]: Yes, location is a part of master data



The Specification tab allows detailed information about how the stock will be managed.
[Stock Type]: The Stock Type entries are held in the Stock Type phrase list and can be updated if required to suit your environment.
[Batch Label]: The Batch Label allows users to maintain the label template so that printing stock label automatically associated the label template.
[Operator & Location]: The Operator and Default Location fields display the 'who' and 'where' for your Stock.

Commented [NR148]: Is target amount is where we will maintain minimum stock quantity? If not, what is this?

Commented [MX149R148]: Tracking Inventory allows you to see how your Stock Batches are being consumed. You will be trying to maintain adequate levels of materials by indicating a target amount (with units) and this information is held on this tab. The Track Inventory box must be selected.

Commented [NR150R148]: But stock consumption will be different based on SOP of materials where this batch is used. It cannot be static, right?

Commented [MX151R148]: It's just for reference amount, not practical amount. So it's static data, and it's optional field to maintain.

Commented [NR152]: What is the use of label here? Is it for finding the stock location, batch mfg and expiry date?

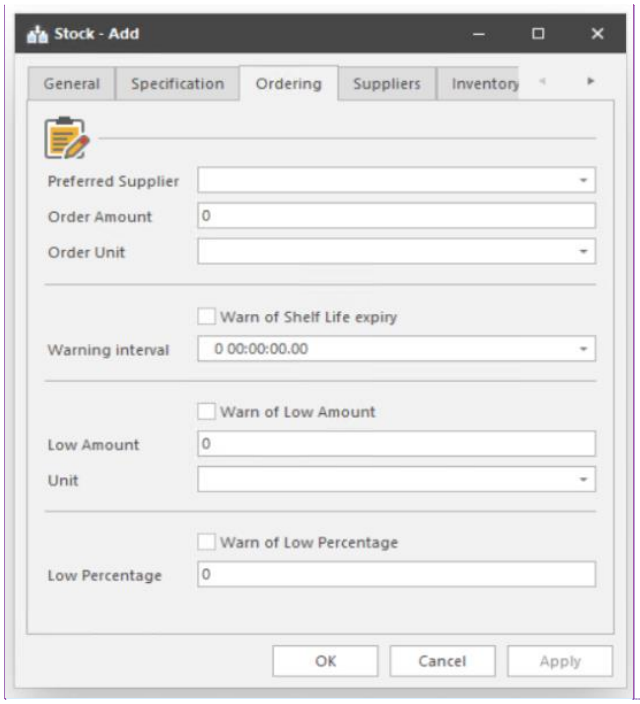
Commented [MX153R152]: It's label template, the information of the label up to you

Commented [HS154]: What does it mean mentioning 'Who'

Commented [MX155R154]: It's field use as stock owner, optional field

Commented [DL156R154]: What is stock owner? Do you mean the modifier of the stock?

Commented [MX157R154]: If you need a responsible people for this stock, you can select here, if not, you needn't maintain this field, it's optional



The Ordering Tab link between Stocks and Suppliers starts here. The record of preferred supplier indicates who you would go to first when ordering Stock Batches. This entry must appear on the Suppliers tab also.

To help automation of the ordering process you can set limits to warn of:

- Shelf life expiry
- Low level in terms of absolute amount
- Low levels in terms of percentage (based on Target Amount)

In Stock interface, will add new fields including handling and storage precautions, security data, associated hazards and risks and GHS requirement.

[Stock Batch]

Sample Management Cabinet → Stock Batch Folder→Right click Add button, will pop up Stock Batch-Add window as shown here. Note: Stock Batch is dynamic data.

Commented [HS158]: The logic of adding stock order should be consumables —— supplier instead of supplier ——consumables.

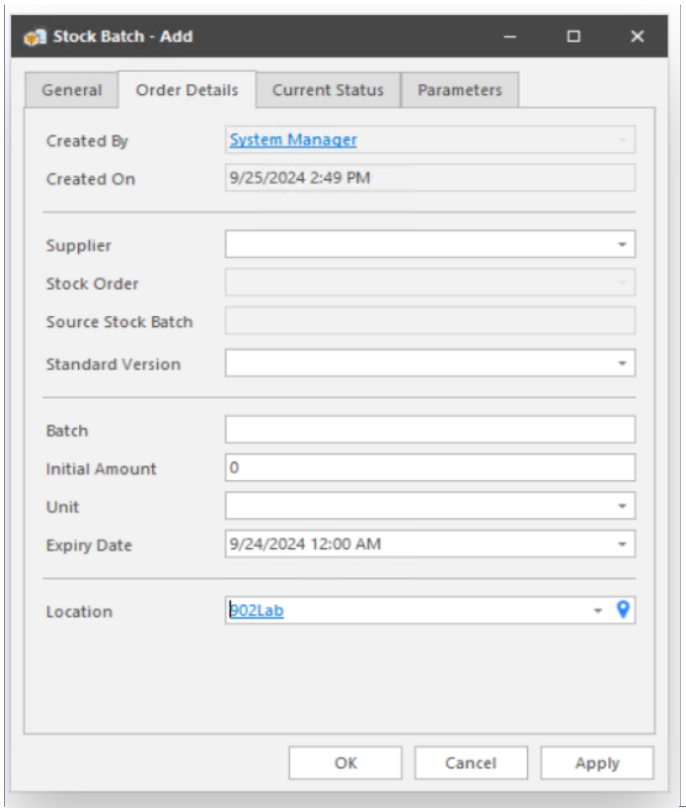
Commented [MX159R158]: It's static data, not ordering operation.

Commented [NR160]: What does ordering stock batches capability in QCLAD perspective?

Commented [MX161R160]: The target amount from the previous tab is copied across to the Order Amount field, but you can change this amount if your ordering processes require this. When you place an order for this Stock (see Stock Management module) the Order Amount value (and unit) is copied from this field automatically

Commented [NR162]: How to manage the stock during production initial day. Will there be a template to upload the stock inventory?

Commented [MX163R162]: First time, we can use template excel to upload stock inventory, and others, you can choose template upload manually or modify in QCLAD manually

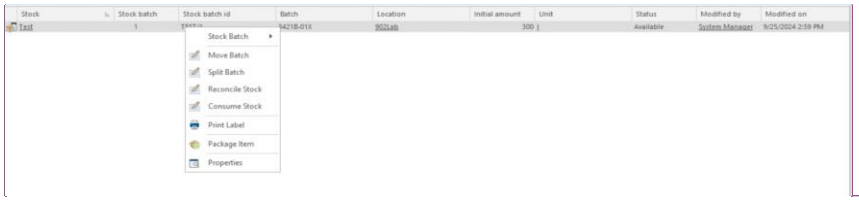


Under Order Details Tab, user can maintain Expiry Date and location for stock, if Stock static data has maintained, these fields will be carried out automatically.

[Consume]

Sample Management Cabinet → Stock Batch Folder→select stock batch and right click Consume button, will pop up Stock Batch-consume stock window as shown here.

(Comments: LES can achieve deduct automatically in PPB. So user can deduct consumption manually if need in PPA.)



Commented [HS164]: Batch of the stock should not be mandatory.

Commented [MX165R164]: Of course, it's optional

Commented [DL166R164]: Please update the FS acccordingly

Commented [MX167R164]: Stock batch is dynamic data, no need to define if it's mandatory.

Commented [HS168]: Stock consume should not managed manually. It should link to test or acitivities. For example, If one test happened, Corresponding amount of consumables should be consumed automatically.

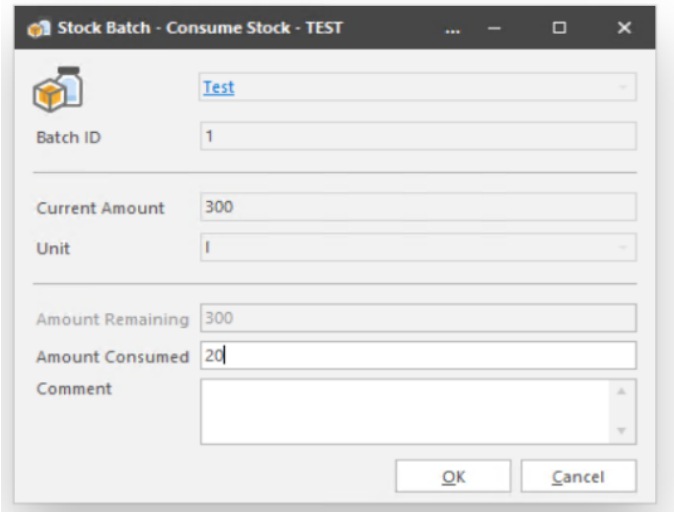
Commented [MX169R168]: In PPA, it needs operation manually, In PPB, it's automatical

Commented [NR170R168]: Is there an alignment for manually performing it will increase the workload for operator?Is it because, this is default pulled from SOP from LES. As LES is not enabled, is it parked for PPB?

Commented [MX171R168]: Yes, exactly, Les can achieve deduct automatically

Commented [DL172R168]: Please update the FS accordingly

Commented [MX173R168]: Done. Add comments here.



~~3.2.4.2~~3.2.5.2 *Instrument Management*

~~3.2.4.2.1~~3.2.5.2.1 *Business Scenarios*

Many laboratory tests are run using instruments and these instruments can be defined in QCLad.

With many instruments of the same type (e.g. gas chromatographs) you will be managing many parts of the same type (e.g. detectors, autosamplers, columns). Records of specific instruments and parts will hold common information based on parameters such as: Location、Responsible operator、Manufacturer、Maintenance: servicing & calibration intervals

In QCLad this common information is held in templates. When a specific item is purchased most of the details for it are defined already and only specific parameters such as identity and serial number need recording.

~~3.2.4.2.2~~3.2.5.2.2 *RTM Mapping*

URS Ref.	Requirement	Type
4.2.1.1.8	System shall be able to display all the instruments (typeA, typeB, type C, typeD) based on the location and method.	Configuration
4.6.1.12.1	System shall allow user to add maintenance plan. Maintenance plan shall include PM(preventive maintenance) type.	Configuration
4.8.1.1.1	System shall support maintain specification and parameters in equipment master data.	Standard

4.8.1.1.2	System shall allow maintain spare parts, calibration and maintenance.	Standard
4.8.1.2.1	System shall support calibration and maintenance plan locally.	Standard
4.8.1.3.1	System shall support to record instrument history, including failures, calibration, maintenance.	Standard
4.8.1.6.2	System shall allow specific user to active expired instrument.	Configuration

~~3.2.4.2.3~~3.2.5.2.3 Function Design

Requirements Description :

Users can maintain instruments information.

Users can view all instruments based on the location and method.

Users can view instruments history.

Design Overviews :

a. Folders and mater menu

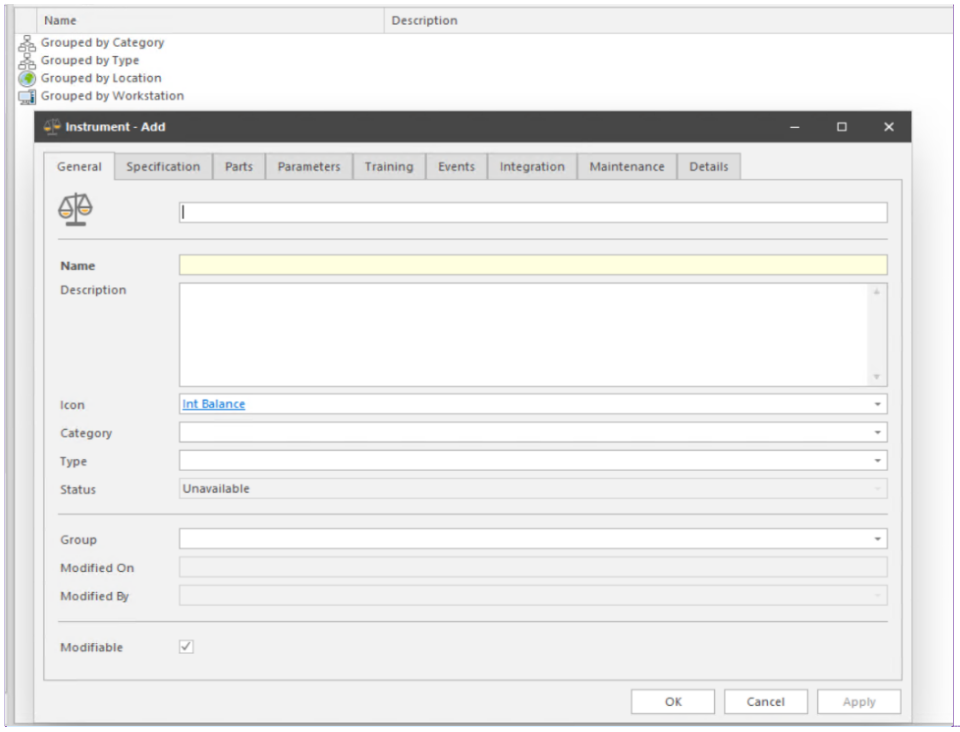
Add Instruments Management Cabinet.

Function Entrance :

Instruments Management Cabinet → Instruments →Right click add button, will pop up the Instrument-Add window as shown here. Users can maintain instruments and parts information and maintenance plan under maintenance tab.

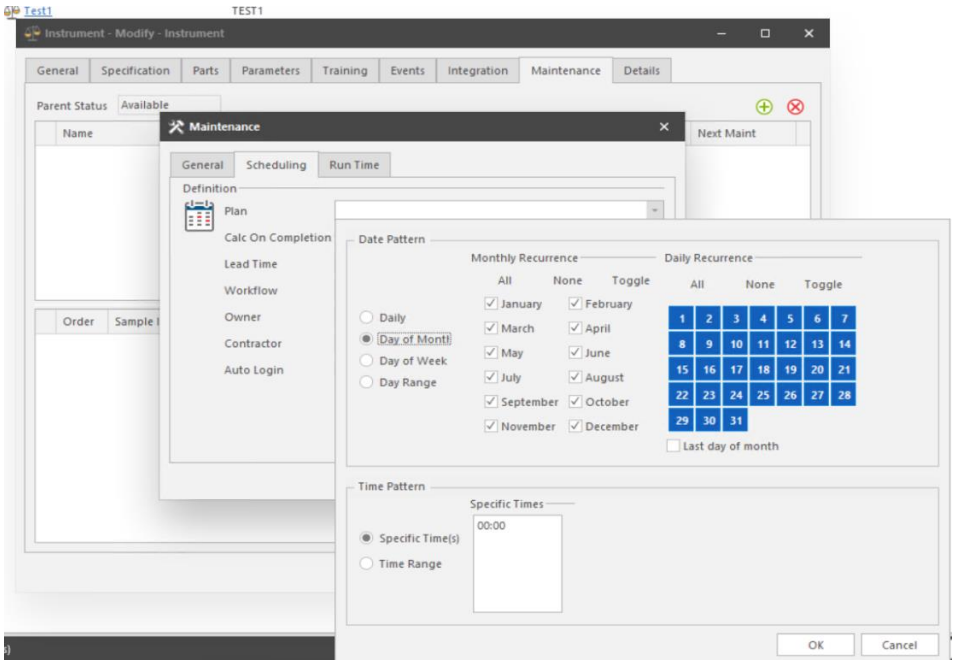
Commented [NR174]: Initial load should be also manual or will there be a provision to mass upload the data and for the new ones we can add manual.

Commented [MX175R174]: Yes, for initial load we use template to upload after collecting master data



Commented [HS176]: This way of maintaining the master data of instrument is not suitable for bulk uploading dozens of instruments. It's better for Thermofisher to provide a excel template for Henkel to key in all instruments and have it uploaded to SM.

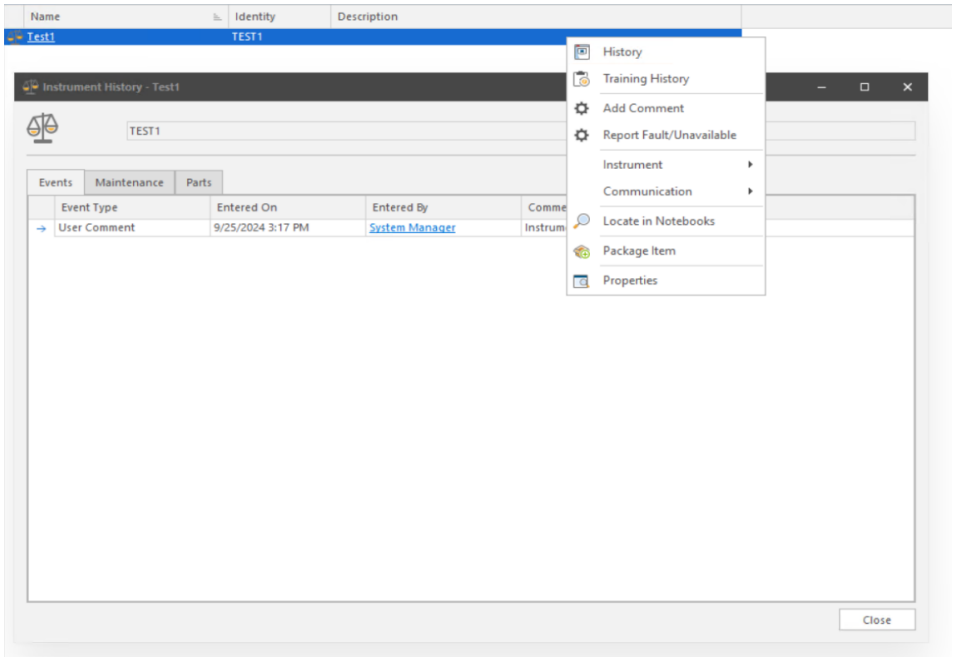
Commented [MX177R176]: Yes, we will provide this when master data collection starts



Instruments Management Cabinet → Instruments →Select specific instrument and then Right click History button, will pop up the Instrument History window as shown here. Users can view instruments history.

Commented [NR178]: What information does it capture?

Commented [MX179R178]: Operation history including operation type, operator, time, comments



~~3.2.4.3~~3.2.5.3 *Label Management*

~~3.2.4.3.1~~3.2.5.3.1 *Business Scenarios*

Labels are used to identify various samples within QCLad. Each type of label needed is created as a Label Template.

Labels will also require a specification to decide on label size and how multiple labels will be printed off.

User want to generate QCLad label for non MES samples and other common samples.

~~3.2.4.3.2~~3.2.5.3.2 *RTM Mapping*

URS Ref.	Requirement	Type
4.1.1.18.3	System shall be able to create labels for sample scan receiving for non MES samples.	Configuration
4.2.1.7.1	System shall allow user to print label by manually input the information.	Configuration

~~3.2.4.3.3~~3.2.5.3.3 Function Design

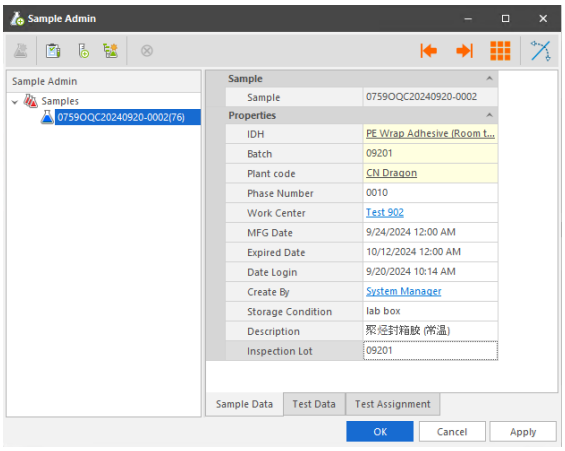
Requirements Description :

Users can manually print label for samples.

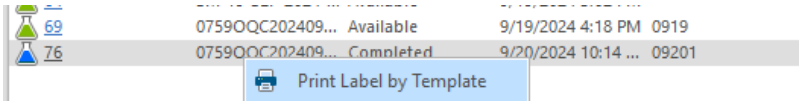
User can enter the label information on samples.

Function Design :

User can enter the information which is shown on label by entering them on sample. Open the sample admin window by clicking the IdNumeric number, enter necessary information in sample data page.



User can print label at any time for samples by right clicking on the sample and select print by label template.



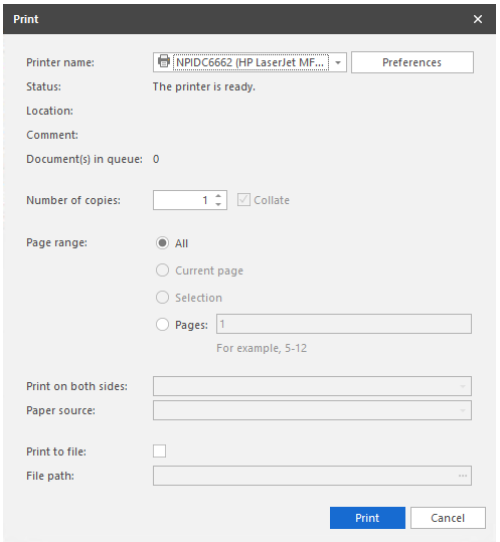
Select label template if needed and then select printer. Click print.

Commented [NR180]: Is storage label template aligned with plant team?

Commented [MX181R180]: No, we only receive three sample label templates

Commented [DL182R180]: What is this particular label for?

Commented [MX183R180]: For printing label, it's standard function for sample or stock you want to print



~~3.2.4.4~~3.2.5.4 *Report Management*

~~3.2.4.4.1~~3.2.5.4.1 *Business Scenarios*

Report Templates are used to generate reports for various entities within QCLad. Each type of Report needed is created as a Report Template. The report template may then be associated with an entity.

Special COA requires a certificate template. A certificate template contains the information required to generate a certificate for sample.

~~3.2.4.4.2~~3.2.5.4.2 *RTM Mapping*

URS Ref.	Requirement	Type
4.1.4.1.1	System shall allow users to define report templates by themselves using reporting tool.	Standard
4.1.4.10.1	System shall support to generate single sample report, multiple sample report, COA report.	Configuration
4.1.4.11.1	System shall be able to route the output of reports to screen, printer, file (CSV, XML, PDF) or email.	Standard
4.1.4.12.1	System shall be able to support various formats (e.g., tabular, matrix, graphical or text) in report templates.	Standard

4.1.4.13.1	Systems performance must not be adversely affected by running reports.	Standard
4.1.4.15.1	System shall allow users to define report templates based on customer or product .	Standard
4.1.4.15.2	System shall allow user to create and modify report templates.	Standard
4.1.4.15.3	System shall allow user to update version for reprot templates.	Standard
4.1.4.2.1	System shall be able to provide editable text style or format in the text report.	Standard
4.1.4.3.1	System must be able to report on the analyzed data at all levels (e.g., raw, recorded and reported analytical data).	Configuration
4.1.4.4.1	System shall collate results from multiple tests for each sample.	Configuration
4.1.4.5.1	System shall be able to support to insert signature picture, company logo, picture, trending chart, table in the reports.	Standard
4.1.4.7.1	System shall be able to provide a library of report templates.	Configuration
4.1.4.8.1	System shall be able to generate analytical reports based on the data recorded in QCLAD like date time, QC operators details along with inspection.	Configuration
4.1.4.9.1	System shall be able to generate and save “Test Report” template.	Configuration
4.1.4.9.2	System shall be able to print test report of other samples based on templates.	Standard

~~3.2.4.4.3~~3.2.5.4.3 Function Design

Requirements Description :

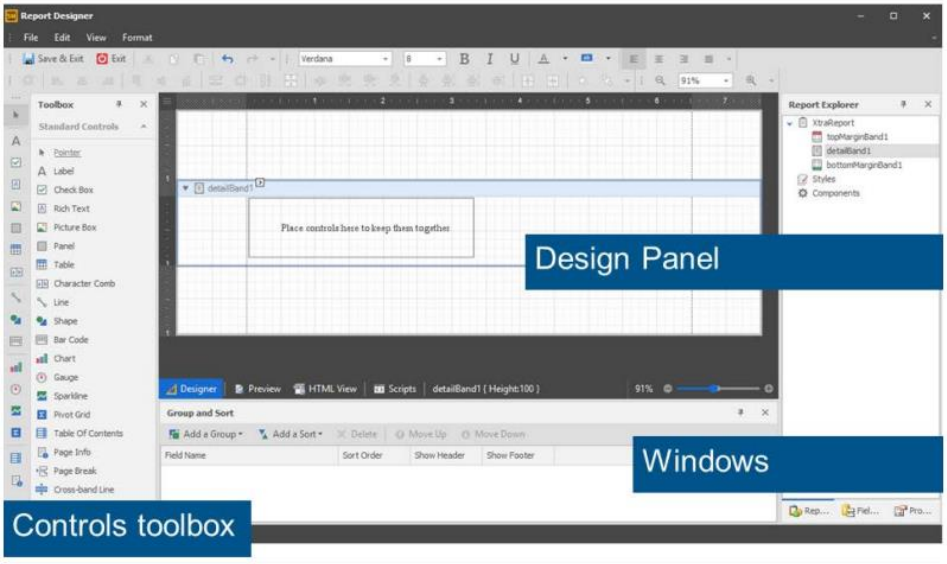
Users can create certificates for single or multiple samples.

User can create report template.

Function Design :

User can create or modify report template in the following path: setup → system → report template → open report templates, open Report Designer in report design page.

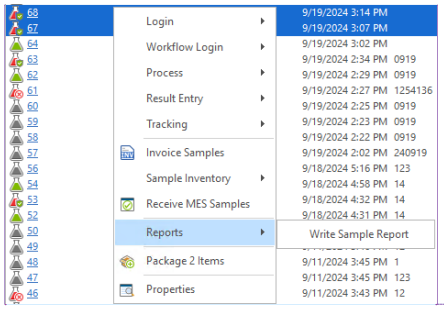
The Report Designer window is divided into areas of interest. Information on each window is contained on the following pages.



User can create certificate or reports by right clicking on authorized samples:

- ***Create single or multiple sample report***

Select sample(s) and right click → reports → sample report to create report.



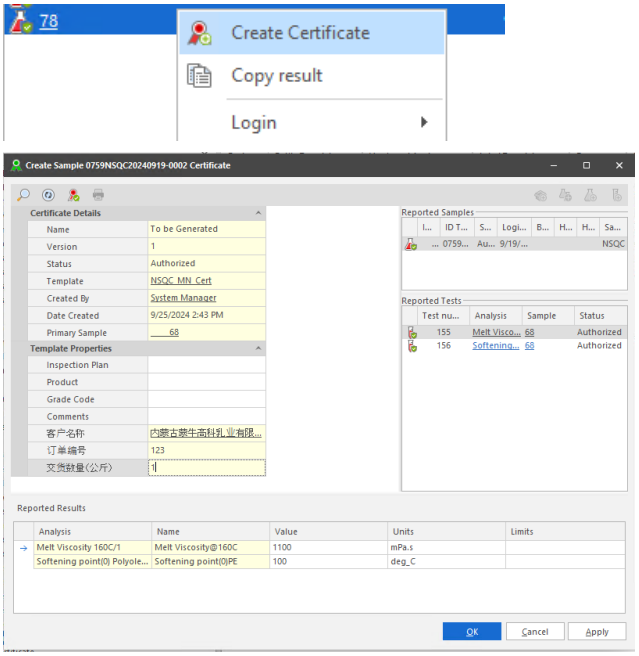
In the preview window, user can print the report or export the report as pdf or docx file.

- ***Create COA certificate***

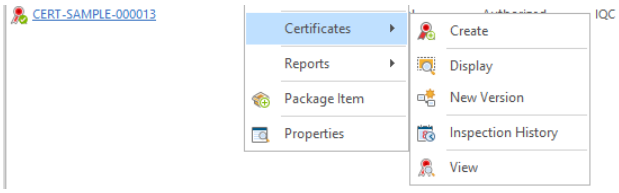
User can create COA (certificates) for authorized samples. Right click on the sample and select 'create certificate'. Select cert template and enter the information needed for the report template and click OK. Certificate will be saved automatically in QCLad.

Commented [HS184]: Will we need to specify which sample go with which template?

Commented [MX185R184]: You can defined on sample type level



User can find the COAs in certificate folder. Detailed information can be found there. User can print or export as pdf, docx file by right click and select 'view' button. User can make a new version of the certificate by clicking 'new version' button.



~~3.2.5~~3.2.6 *Interface*

~~3.2.5.1~~3.2.6.1 *General requirement*

~~3.2.5.1.1~~3.2.6.1.1 *Interface configuration*

For the interface, such as QCLAD to SAP inspection result, each interface can maintain its own configuration item with below information.

Table Field Name	Comment
Identity	The ID of the outbound interface
Email	The address to which an email will be sent when any exception or error occurs when processing this interface.
Description	Plain text field
Type	Inbound or outbound
url	**Will be only applicable for outbound interface:** The SAP endpoint to which the json from QCLAD will post
Request header	**Will be only applicable for outbound interface:** One or more request header parameters required by SAP can be defined in this tab. Please refer to below screenshot

Request header reference screenshot:

Interface Headers

Header Name	Header Value
service_id	Sample
service_name	Sample
source_sys_id	LIMS
submit_time	
target_sys_id	SAP
trans_id	<SYSTEM>
username	system

~~3.2.5.1.2~~3.2.6.1.2 *Interface log*

For each interface transaction, whether inbound or outbound, one interface log will be correspondingly created to record below information.

Table Field Name	Comment
Identity	The ID of each interface log
Interface name	The name of the interface
Interface type	Either inbound or outbound
Transaction ID (UUID)	For the inbound interface, the transaction ID is populated from the UUID given in the SAP json. For the outbound interface, the transaction ID is auto populated with below naming convention:

	Interface Name + current day (YYYYMMDD) where the log is created+ 4 digit incremental number.
Created date	The date the log is created
Response code	Only applicable for outbound interface i.e. 200, 400, 500 etc
Status code	The processing outcome Success or Failure
Status message	The processing outcome
Request body	The request body json will be recorded as a blob value.
Response	The response json will be recorded here as a blob value.
Process log	The processing outcome log will be recorded here a blob value. the processing outcome log will include below components. 1) The timepoint of start processing 2) The entity ID (s) is being processed 3) The processing outcome 4) The time point of end processing 5) The time point of interface log creating

3.2.6.1.3 Exception reporting

One json transaction can contains more than one objects. For example, one json of inspection lot interface can have more than one inspection lot(s), thus, system should process and commit each objects within one json respectively and independently.

For instance, if one json contains 3 inspection lot, and 2 of 3 lots are error free, then system should save these 2 lots into database, and report error on the other one.

And for each object, system should identify all the errors and collectively report back to SAP. For example, for one plant (one object) within one json, there are two erroneous fields, then QCLAD should identify all and report back all the errors.

And the solutional action require SAP to push again, SAP will only push the erroneous object down to QCLAD.

~~3.2.5.1.3~~3.2.6.1.4 Exception handling

Email notification:

For the inbound interface, if the transaction from SAP haven't reached to QCLAD, due to reasons such as internet inaccessibility or power outages in QCLAD end or otherwise, SAP should follow its mechanism to process such kind of exception.

However, if the transaction does reach to QCLAD and any error occurs during processing, along with log creation, if the any error, namely the status code being "failure", QCLAD will query the email address (could be a DL email

address) maintained in current subjected interface, and send out an email to that address containing below information:

On the email subject: Interface name + current date + "Interface error" (being a fixed text)

On the email body:

1) Interface name
2) Transaction ID
3) Date of error
4) Error message

Error and Actions knowledge:

Along the way during implementation and system go-live, QCLAD should maintain a MS word document to collect all the errors and their corresponding solutional actions.

~~4)~~ Before system go-live, this document will be primarily maintained by project team; after go-live, it is by System Administration team.

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Re-processing of transaction

Upon receiving such email as interface alert, the responsible person can make necessary actions with accordance to the error message, and then if needed, can manually locate and right click on the subjected interface log to reprocess the transaction.

For example, an incoming transaction of material master data is for plant A; however, plant A is not yet created in QCLAD, thus fail to successfully process this transaction. Hence upon receiving the email notification, the responsible person can communicate with SAP to push the plant A master data down to QCLAD; and subsequently either ask SAP to repush the material master data transaction again or manually reprocess the log in QCLAD end.

~~3.2.5.2~~3.2.6.2 RTM Mapping

URS Ref.	Requirement	Type
4.5.1.1	System shall support download inspection lot from SAP to QCLAD. 1）Create a sample in QCLAD based on the downloaded Inspection Lot, or change an existing sample (JIT process). 2）If the inspection lot in SAP also has multiple activities, then QCLAD needs to create multiple samples. 3）The test item in lot has a field for whether or not to skip the batch (dynamic message), and whether the SMIDI standard function can be written directly to the test table.	Interface
4.5.1.3	System shall support download Inspection Plan from SAP to QCLAD. Needs to be synchronised and updated in real time.	Interface
4.5.1.4	The system shall support the downloading of Inspection plans from SAP to QCLAD whose Inspection Plan contains the MIC. Needs to be synchronised and updated in real time.	Interface
4.5.1.5	The system shall support the downloading of Inspection plans from SAP to QCLAD whose Inspection Plan contains the specification. Needs to be synchronised and updated in real time.	Interface

URS Ref.	Requirement	Type
4.5.1.6	The system shall support the downloading of Inspection plans from SAP to QCLAD whose Inspection Plan contains the Material master data. Needs to be synchronised and updated in real time.	Interface
4.5.1.8	System shall support download Plant Code from SAP to QCLAD.	Interface
4.5.1.9	System shall support download QM notification from SAP to QCLAD.	Interface
4.5.1.10	System shall support to upload result value from QCLAD to SAP. 1） When there is a Retest operation done, the result of the last Retest needs to be returned to SAP. 2） once the LIMS post result back to SAP， SAP MIC status should be auto rejected or accepted? 3） system shall support post all the samples result back to SAP, including the resampled sample. 4） related to item 1, whether Smidi can configure that, once inspection lot is downloaded mapped with an existing sample in LIMS; if that sample is authorized with result, then Smidi can also trigger the result pushing to SAP? (JIT process)	Interface

3.2.6.3 SAP to QCLAD Plant Master data

3.2.6.3.1 Introduction

The plant master data is currently maintained in SAP to represent each factory site across the global. The plant ID is unique and not duplicate among SAP P92 and P93. And under each plant, there are multiple work centers which is maintained to represent the laboratories.

Any changes made on plant master data or work center master data, once approved, will trigger to push down to QCLAD via a scheduled job on SAP with configured frequency to ~~real-time~~ synchronize the data between SAP and QCLAD.

SAP will post the data via API down to QCLAD.

3.2.6.3.2 Interface URL address

https://{*QCLAD domain name*}:56105~~0~~/SAPtoLIMS/MasterData/PlantAndWorkCenter

Notes: the full URL address is up for changes, above is for reference.

3.2.6.3.3 Field Mapping

Refers to the execl file sent along with email confirmation.~~Refers to below embedded Excel files.~~

~~Note: the LIMS's field name will be left empty since the database table or fields is not yet created. The database table and fields along with its properties will be created accordingly based on the understanding of what is coming from SAP.~~

Commented [NR186]: Not real time only on scheduled job daily once or on a configured interval agreed by both the parties.

Commented [XW187R186]: Either way is fine as long as the frequency of data sync-up between SAP and QCLAD is accepted by business side.

Updated the description accordingly.

Commented [ZZ188R186]: If this is done once a day or at configured intervals, is the Plant data sent from SAP to QCLAD only the data that has been changed since the last time it was sent to the current time, or all of it?

Commented [XW189R186]: After system go-live, every job triggering will be triggering only the incremented data.

3.2.6.3.4 Exampled transaction Json

Refers to ~~below embedded TXT files.~~the json file sent along with email confirmation.

Plant Master Json.txt

Field Code Changed

3.2.6.3.5 Exampled response json

Please find below exampled response json:


```
{
  "InterfaceCode": "HEN_PLANT",
  "Code": "1",
  "ProcessOutcome": [
    {
      "TransactionID": "UUID1",
      "Status":"0",
      "Message": "Completed Processing PLANT Data A "
    },
    {
      "TransactionID": "UUID2",
      "Status":"1",
      "Message": "Error message 1 | Error Message 2 for plant Data B "
    },
    {
      "TransactionID": "UUID3",
      "Status":"0",
      "Message": "Completed Processing PLANT Data C "
    }
  ]
}
```


~~{~~

~~"InterfaceCode": "HEN_PLANT",~~

~~"Code": "1",~~

~~"Message": "Completed Processing CUSTOMERPLANT Data "~~

~~"TransactionID":"xxxxx"~~

Commented [NR190]: What do you mean by customer data? Is it an illustration?

Commented [XW191R190]: It is redundant description. will delete it.

}

Where

"InterfaceCode" : ~~field~~ would provide the name of the interface;

"Code" : will have value 1 if all~~for~~ success and 0 ~~for~~if any failure;

"~~Message~~ProcessOutcome": ~~will provide more information of processing outcome~~is an object array to provide all the process outcome for each object within one json.

"TransactionID": is the UUID of ~~each transaction~~each object within one json for reference.

"Status": will have value 1 if current object is success and 0 if failure;

"Message": is the process outcome. If not then one error, will be separated using "|".

3.2.6.3.6 Frequency

~~Real-time~~on a scheduled job with a configurable frequency in SAP. ~~Whatever change made on the plant master data or work center~~whenever the job is running as per frequency, SAP will push the all changes to QCLAD.

3.2.6.3.7 System Logic

Below is to detail on the further logic which is not included in the field mapping file when QCLAD processing the incoming json.

1) Upon receiving the transaction, QCLAD will either newly create or updating existing plant master data and work center master accordingly.
2) Below information will be auto created for each plant master data.

Table Field Name	Comment
Identity (Plant_code)	Unique ID field Detailed mapping can be found in excel file.
Name	Detailed mapping can be found in excel file.
Source_system	Links to a phrase (optionlist), have two options: P92 and P93
If_warehouse	Boolean field
Parent_plant	Linking to plant master data
Customer	Plain text field
Vendor	Plain text field
Alias_name	Plain text field
Address	Plain text field i.e.: streetHno,city,countryTXT,Countrykey

Commented [NR192]: Is interface code plant code? We will be using the same structure for incremental data. In that there can be data of multiple plants as we onboard more plants.

Commented [ZZ193R192]: interface code refers to the interface code in QCLAD, independent of the factory.

Commented [NR194]: Do you have a list of messages which can be shared so that both parties are aware what messages can be expected and what does each message mean?

Commented [XW195R194]: We can only provide the a few typical messages example and its syntax to you.

This message field is a comment response of the processing outcome.
Say, if all success, will response: xxx data is successfully created.
However, if failed at any code, the error message will vary, hence, it is unable for us to predict and provide all the error message.

Commented [XW196R194]: Lelsie will have one more section to define the exception/error handling process and functionality.

Commented [NR197]: Master data incremental change is not real time, it is daily based on a scheduled job.

Commented [XW198R197]: As above,
Updated the description.

Commented [NR199]: The warehouse means some inspection lots of Dragon is created within warehouse which should also be considered in QCLAD. It should be displayed as parent plant's warehouses.

Commented [XW200R199]: As understood from the excel, there are two fields provided by SAP.
1: if warehosue: it yes, meaning current plant master data is actually represent a warehouse. Hence, needs to further understand to which physical plant the warehouse is situated, which is maintained in the second field.
2: parent_plant: hence, this field should be linked to a plant master data.

3） Below information will be auto created for each work center master data. Be noted that one json can contain more than one work center data under the child node.

Table Field Name	Comment
Identity	Auto concatenated by plant code (werks) +” ”+work Center name (workCenter) to make its uniqueness.
Work_center	Detailed mapping can be found in excel file.
Plant_code	Links to the plant code to build the linking between current work center and plant.
Source_system	Links to a phrase (optionlist), have two options: P92 and P93
category	~~Links to a phrase (optionlist),~~plain text field
usage	Plain text field

4） User with privilege can manually create, modify, display, remove/restore the plant or work center master data in QCLAD.

5） Users can only view plant or work centre master data in QCLAD.

Commented [NR201]: Creation, modification of work center master data should not be allowed as it need to be managed in SAP only

Commented [XW202R201]: The maser menu will be created in LIMS.
However, in the configuration, those master menu will not be assigned to anyone, so that no one is able to manually create it in PROD.

Formatted: Outline numbered + Level: 4 + Numbering Style: 1, 2, 3, … + Start at: 1 + Alignment: Left + Aligned at: 1.77" + Tab after: 2.37" + Indent at: 2.37", Tab stops: 2.91", List tab

~~3.2.5.1~~3.2.6.4 SAP to QCLAD Material Master Data

~~3.2.5.1.1~~3.2.6.4.1 Introduction

SAP is the source of data to maintain material master data. Each material code is unique across two Henkel SAP system. One material code can be assigned to more than one plant and links with multiple inspection types.

One particular points is that currently SAP haven’t maintain and standardize the production line and assign to each material code. Hence, upon material is pushed and created in QCLAD, QC personnel needs to manually assign one production line to the material code if needed.

Material master will be pushed to QCLAD as per the frequency defined in the scheduled job in SAP.

~~3.2.5.1.2~~3.2.6.4.2 Interface URL address

https://{*QCLAD domain name*}:56100/SAPtoLIMS/MasterData/PlantAndWorkCenter

Notes: the full URL address is up for changes, above is for reference.

~~3.2.5.1.3~~3.2.6.4.3 Field Mapping

Please refer to attached excel file sent along with confirmation email.

Commented [NR203]: Which one to refer to?

Commented [XW204R203]: For now, you can refer to the excel file in sharepoint.

In the final approved version, I would sent out the independent clean-version excel.

~~3.2.5.1.4~~3.2.6.4.4 Exampled transaction json

Please refer to json file sent along with confirmation email.

~~3.2.5.1.5~~3.2.6.4.5 Exampled response json

Please refer to chapter 3.2.6.4.5 Exampled response json

~~3.2.5.1.6~~3.2.6.4.6 Frequency

As frequency defined in the scheduled job in SAP.

~~3.2.5.1.7~~3.2.6.4.7 System Logic

1) Upon receiving the transaction, QCLAD will either newly create or update the existing material master data. The material master data is broken down into the plant material level with below information:

SAP field Name	SAP Field description	Table Field Name	Comment
Src_system	Source system	source_system	Links to a phrase, having options P92 and P93; this value should be replicated into each material objects within current json.
-	-	identity	Concatenated the plant code(WERKS) +"_"+material code (MATNR)
WERKS	Plant code	plant	link to plant master
MATNR	Material IDH	code	plain text field
MAKTX	Material Description	name	plain text field, material name which is recognizable by end user.
BKLAS	Valuation class	type	Links to a phrase, the possible values are as below: ROH: raw material ~~HAW~~ROHV: packaging HAW: Traded goods UNF - Semi finished goods, FER - finished goods
DISPO	MRP controller code	category	plain text field grouping of material, internally for the plant reference.
MMSTA	Plant material status	status	Plain text field. As Dragon not strictly follow the rule, save it as reference for the time being.
XCHAR	Batch managed or not	batch_management	boolean field X means TRUE NULL mean FLASE

Commented [NR205]: Updated valuation class as there was a mistake

Commented [XW206R205]: Noted.

PRFRQ	Inspection interval	Interval & Interval_unit	Real & links to phrase (Days/ Month/Year) The default unit is “Day”. recurring inspection lot creation will be auto created considering this value. this logic will be followed and used in SAP.
~~PSTAT~~		~~???~~	~~plain text field~~
LVORM	Flag for deletion	remove_flag	boolean field X means TRUE NULL mean FLASE
MHDHB	Shelf life	shelf_life & shelf_life_unit	Real & links to phrase (Days/ Month/Year) retention sample: production date + shelf life; more for packakging material.
LGTYP	storage type	Tank_material	Boolean field If given value is “TAK”, then TRUE If else, then FALSE
ART	InspectionType	inspection_type	plain text field Raw material: 01PQ: IQC. 09P: recruing inspection lot 89P: trail inspection lot 04*: semi-finished or finsiehd production inspection for now, QCLAD will just store it as reference information.
AKTIV	Active	Active	boolean field X means TRUE NULL mean FLASE
APA	Prefered inspection type	if_prefered	boolean field X means TRUE NULL mean FLASE

Commented [XW207]: What is the purpose of this field?

Commented [NR208R207]: You can omit it

Commented [XW209R207]: Noted.

DYNREGEL	dynamic modification rule	Modification_rule	Plain text field for now, QCLAD will just store it as reference information.
MPDAU	average inspection duration	duration & duration_unit	Real & links to phrase (Days/ Month/Year) SAP auto calculates inspection start date + this duration to calculate inspection end date for now, QCLAD will just store it as reference information.

~~3.2.5.2~~3.2.6.5 SAP to QCLAD Batch Information Transaction Data

~~3.2.5.2.1~~3.2.6.5.1 Introduction

Batch information is a transaction data maintained in SAP to document the batch number, production date and expiry date etc information of each batch for the raw/packaging materials or finished product.

Particularly with semi finished products and finished products, process order will link with a batch information to initiate the production, and thereby to trigger the inspection lot in SAP. The inspection lot at this point will be pushed to QCLAD with the Batch No.

Commented [NR210]: Applicable to semi finished and finished goods.

Commented [XW211R210]: Noted and updated.

However at this point, the production date and expiry date is not yet available in the batch information and thus not able to come along with inspection lot down to QCLAD. These two value will be available only after production is all completed and posted into warehouse. And this two value is needed in retention sample created out of finished product to keep track of its expiry date.

Hence, when production is completed and posed in SAP, the batch information will be updated with production /expiry date in SAP and pushed to QCLAD. QCLAD should subsequently query the inspection sample and retention sample, whose material, plant and batch no is the same, and update the production / expiry date into them.

~~3.2.5.2.2~~3.2.6.5.2 Interface URL address

https://{QCLAD domain name}:56105/SAPtoLIMS/TransactionData/BatchInformation

Notes: the full URL address is up for changes, above is for reference.

~~3.2.5.2.3~~3.2.6.5.3 Field Mapping

Please refer to json file sent along with confirmation email.

~~Refers to~~

~~3.2.5.2.5~~3.2.6.5.4 Exampled transaction json

Please refer to json file sent along with confirmation email.

~~Refers to~~

~~3.2.5.2.7~~3.2.6.5.5 Exampled response json

Please refer to chapter 3.2.6.4.5 Exampled response json

~~3.2.5.2.8~~3.2.6.5.6 Frequency

As frequency defined in the scheduled job in SAP.

~~3.2.5.2.9~~3.2.6.5.7 System Logic

1) ~~However,~~ when SAP update the production and expiry date on any batch information and push to QCLAD, QCLAD will query the inspection samples and retention sample whose plant, material and batch no is the same as current batch information, and then update the production /expiry date into the corresponding fields of those queried samples.

~~3.2.5.3~~3.2.6.6 SAP to QCLAD Inspection Plan Master Data

~~3.2.5.3.1~~3.2.6.6.1 Introduction

Each material (IDH) has its own inspection plan created for it if inspection is required in SAP. One inspection plan can contain more than one activity no. for instance, a SFG or FG material would have one inspection plan which contains two activities No to further define the test parameters (MIC) and its spec respectively for IPQC and OQC. And each activity no can contain more than one test parameters (MIC). And each MIC will be linked with Test method, has its own short text (the reported name), has info field 3 to further define the testing conditions which is either included in the CoA reporting to help to further differentiate the testing points.

Currently, SAP haven't standardized the Test method and MIC code.

Regarding the test method master data, it is not strictly used to represent the offline SOP. For example, the same Test Method can be used in different inspection plans yet represent different analytical procedures. Henkel decides to standardize the test method master data on factory level, to make sure its representation is corresponding uniquely to one offline SOP and its usage is consistent across all the inspection plan with factory.

Regarding the MIC code, the same combination (MIC code + test method) could be added more than once in one inspection plan along with different info field 3 to indicate different testing parameters; and furthermore, the same testing parameters, if in different inspection plan, are used with different combinations (test method + MIC code). Since MIC is too critical to be standardized and updated directly, it is decided to have additional field to maintain the standardized MIC code for the type B MIC record in inspection plan. For those non-COA and non-Type B MIC, with current understanding and alignment, current MIC code will be used as the unique key for each test method.

One inspection plan/ activity no itself contain the test method, and the testing parameter and the spec of each testing parameter when a material is inspected. Correspondingly, when interfaced down to QCLAD, due to data structure differences, QCLAD altogether needs to create below master data:

~~3.2.5.3.2~~3.2.6.6.2 Interface URL address

https://{QCLAD domain name}:56105/SAPtoLIMS/MasterData/InspectionPlan

Commented [NR212]: IPQC and OQC mostly come in bulk and there can be FG also

Commented [XW213R212]: Updated the example in which the material type is changed to SFG (bulk)

Commented [NR214R212]: It can be SFG, FG, so please add both

Commented [XW215R212]: added

Commented [NR216]: This will be populated strictly only for those MICs where results will be pulled for Type B equipment.

Commented [XW217R216]: We are expecting, for Non-COA MIC, regardless of type B MIC or Non-Type B MIC, SAP end should maintain a value under this additional field (unique MIC field) to ensure the consistency.
Under this case, from SAP end, if it is type B MIC, then you maintain this offline aligned code; if it is non type B MIC, then you can current MIC code into this field.

Otherwise, when processing the inspection plan, QCLAD needs to have three sets of logics(COA MIC, Type B MIC, non-Type B MIC), that is lots of work and not good for interface performance. And this is not we agreed.

Commented [NR218R216]: This needs discussion. Can't you do that copy and not sure what is the use of it. Type B we are giving, Check CoA you are differentiating with inspection method and all others are MIC ID as we are not doing any such checks are at our end.

Commented [XW219R216]: We can't think of a way to consolidate the all the scenarios into two sets of logic only.

Because QCLAD has to start from Test Method (Analysis) level not component level.

Commented [XW220R216]: Updated the description as aligned on the logic.

Notes: the full URL address is up for changes, above is for reference.

~~3.2.5.3.3~~3.2.6.6.3 Field Mapping

Please refer to json file sent along with confirmation email.

~~Refers to~~

~~3.2.5.3.5~~3.2.6.6.4 Exampled transaction json

Please refer to json file sent along with confirmation email.

~~Refers to~~

~~3.2.5.3.7~~3.2.6.6.5 Exampled response json

Please refer to chapter 3.2.6.4.5 Exampled response json

~~3.2.5.3.8~~3.2.6.6.6 Frequency

As frequency defined in the scheduled job in SAP.

~~3.2.5.3.9~~3.2.6.6.7 System Logic

1. **If process the inspection plan json and create master data in QCLAD accordingly:**

- As long as the combination of json field “VERWE” (Usage) and “STATU” (Status) not have the value being “5” and “4” respectively; then it would be understood as current version of inspection plan is changed and disabled. SAP would filter pushing such kind of transaction down to QCLAD. However does pushed to QCLAD, QCLAD not need to process such kind of transaction json.

2. **Based on understanding of SAP inspection plan master data, the Test Method (Inspection method) is corresponding to Analysis master data in QCLAD while MIC to component in QCLAD. In QCLAD, one Analysis can have more than one MIC. Hence, when processing each MIC object, QCLAD needs to create the Analysis and component master data using below further detailed requirement.**

- firstly, QCLAD needs to determine if current MIC object is a COA MIC or non-COA MIC. if the inspection method ID (pmethode) is equal to "XXXX", it would be seen as COA MIC; then follow point 3 for further requirements; if not equal to "XXXX", it would be seen as non-COA MIC, then follow point 4 for further requirements.

3. **Analysis and component master data creation for each COA MIC**

- Currently, Henkel doesn't manage inspection method for the COA MIC. Basically, it is decided one COA MIC will be created with one Analysis. Hence, QCLAD should base on the combination of **“plant code” + “MIC Code” + ”Short Text”** to determine either to update or newly create an Analysis.
- If need to newly create an Analysis, the naming convention to **automate the Identity (ID)** of Analysis is "C+9 incremental digit number"; and the naming convention to **automate the Name** of Analysis is "MIC Code+short text".
- And regarding component creation under this Analysis, QCLAD will take the **Short text** as component name (the unique key of component), and save the order no, unit, decimal place, short text, mic code, info field 3 respectively along into each MIC (component), so that at the time when creating virtual sample, if this Analysis is used, these information can be displayed on Result entry.

Commented [NR221]: For Dragon the inspection method is confirmed as LOMCOA01 for check CoA. This implies only for raw materials. Will there be a configuration to define this to avoid any hardcode at plant level

Commented [XW222R221]: Okay, noted. We do it this way.
Will add a field on plant master data where user can configure the Test Method for COA MIC.
Hence, after a new plant is created, user needs manually maintain this information.

The reason to have this configuration on each plant master data is because maybe each plant might have different COA test method id.

Please confirm it is accepted?

Commented [NR223R221]: It has to be there in your master data list (mandatory)

Commented [XW224R221]: Yes, we can add a field and make it mandatory in plant master data.

What I want to emphasize is that, since plant master data is originally created by SAP and without this information,

Hence, after its created from interface, user needs to maintain such value once.

It is not a huge work.

Commented [NR225R221]: Need to provide the details in handing over document

Commented [XW226R221]: Looks like it is some sort of deliverable document.
Please let Wendy know and get aligned on what is really needed.

Commented [NR227]: Hope this is inspection characteristic number, identifier of each row in SAP inspection plan.

Commented [XW228R227]: Yes, this field is of that purpose.

- Particularly, **regarding the type of component**, If equal to “QUAN-01”, then set the result type as “N” (numeric); If equal to “QUAL-01”, then set the result type as “O”(option), and links to a phase that has below two options :

  0: Pass

  1: Fail

- Lastly, no need to create component list for this kind of Analysis. Since this kind of Analysis only has one component for it.

Commented [NR229]: @Ashu Singh (ext) can you please confirm whether from QUAL what results are you expecting from QCLAD

4. **Analysis and component master data creation for each non-COA MIC**

- Currently, Henkel decides to standardize the inspection method on plant level. Basically, all the non-COA MIC will be assigned with an individual Test method. And Test method is one to one relation to offline SOP. Hence, QCLAD should base on the combination of **"plant code" + "Test method"** to determine either to update or newly create an Analysis master data in QCLAD.
- If need to newly create an Analysis, the naming convention to **automate the Identity (ID)** of Analysis is "TM+8 incremental digit number"; and the naming convention to **automate the Name** of Analysis is "plant code+Test Method Name (a value provided in interface)".
- And regarding component master data creation under this Analysis, QCLAD will take the **unique MIC code** (the field provided by interface) as component name, these kind of MIC are so called Type B MIC;if the unique MIC code doesn’t have any value, QCALD will fetch the SAP MIC code as the component name, these kind of MIC are so called non-COA and non type B MIC . meaning, for two MIC, if they link to the same test method and share the same unique MIC code, then they would merge as one component for that Analysis in QCLAD.

The variance of "order no, short text and mic code, info field 3" will be defined in the Component list level. The detail can be found in following point.

Note1 : and QCLAD will only differentiate the COA MIC and Non-COA MIC. Yet won’t be able to further differentiate the Type A MIC and Type B MIC under non-COA MIC. Hence, for all non-COA MIC, SAP ends needs to maintain the unique MIC code.

- At the time of creating the each component, the decimal place and unit would be defined on the component level.
- Particularly, **regarding the type of component**, If equal to “QUAN-01”, then set the result type as “N” (numeric); If equal to “QUAL-01”, then set the result type as “O”(option), and links to a phase that has below two options :

0: Pass

1: Fail

Note 1: if needed, user need to modify the Analysis master data created to add the intermediate components or update the result type from N (Numeric) to K (Calculated) and configure the formula according.

In development, if subjected MIC is already existed, yet should not update the result type.

Commented [NR230]: Except Type B and RM non CoA. For Type B MIC, it should be unique ID created as per earlier discussion

Commented [XW231R230]: As answered above.
TF only agreed to have two sets of logic to process the inspection plan, namely the COA MIC and non-COA MIC.

However for Non-COA MIC, SAP needs to populate a value under this additional field.

Commented [XW232R230]: Update the description on this part as well.

Commented [NR233]: This will be taken in the initial testing. In case there are any issues, we may need to change in future as Henkel cannot guarantee the outcomes of using MIC ID as component. Also TF has confirmed that for result entry screen and pushing the results to SAP, this will not be constrain.

Commented [XW234R233]: Firstly, for non Type B MIC, as per discussion, it is initially understood from Henkel that, for each Test method, current MIC code used across all the inspection plan should be unique.
Hence, it is agreed between Henkel and TF, that SAP can populate current SAP MIC code into the additional field where the standardized MIC code is maintianed, and then QCLAD will use this value as the unique component name for each Test Method.

Only under this assumption, TF can confirm for result entry screen and result sending back.

However in the future if any otherwise case, it should be respectively discussed and see if current developed logic can be accommodating. TF won’t guarantee the functionalities for all the current SAP data usage that we are not completed informed.

Commented [AS235]: Note: we are ok to get results as pass = ‘0’
Fail = ‘1’

Commented [XW236R235]: cool

Note 2: Dragon plant has types of "QUAN-01" and "QAUL-01"; however, other factory might have value such as "QUAN-05", which means that the result type is quantitative, however needs to replicate the MIC 5 times to allow key in results and all send back to SAP. For PPA, this logic will be not included.

- Firstly, due to each component (MIC) of one Analysis (Test Method), its MIC code, short text and info field 3 can be different among different inspection plan. And secondly, for different inspection plan, even one test method is inspected, however, the number of MICs to be inspected and recorded can be different. Hence, QCLAD needs to accommodate such variance on each Analysis by using Component list. And **one component lists is maintained for the breakdown of "IDH + Activity No" for each Analysis**. So one component list of an Analysis can maintain the actual number of MICs to be inspected, and their order no, MIC code, short text and info field 3 can be respectively defined. So at the time of sample registration, if the subjected component list is used

- At the time of creating component list, the naming convention to **automate the Identity** is "IDH code+_+ Activity No" for differentiation.

5. **One IDH can have only one inspection plan on each plant. One inspection plan can have more than one Activity No. and one Activity No can contain more than one MICs. And Each MIC is defined with specification. In order to build the master data accordingly, one inspection plan will be created with Test Schedule and MLP in QCLAD. Whereas Test schedule is to define what are the Test methods needs to be inspected if used. And MLP is where the specification is defined for all those Test Method and their component.**

6. **Creation of Test Schedule master data, the breakdown of Test Schedule should be combination of "Plant code" +"IDH code" + "Activity No". So that, at the time of processing inspection lot, QCLAD can base on these three factor to match the test schedule and assign to the same to create all the test.**

7. **Creation of MLP master data , the breakdown of MLP should be combination of "Plant code" +"IDH code" + "Activity No". So that, at the time of processing inspection lot, QCLAD can base on these three factor to match the MLP and assign to sample.**

8. **Since each MIC in SAP maximumly can have two sets of specification, namely the internal control and customer specification.**

10. **Each inspection lot is always assigned with the latest active inspection plan at the time in SAP, hence, QCLAD needs to keep the historical records of inspection plan processed and stored in QCLAD and ensure each inspection lot should be linked with all the master data created out of inspection plan.**

Specifically when creating MLP master data, QCLAD needs to have version management. if already existed, then a new version of MLP should be created, and deactivate the prior version.

Since Test Schedule has no version management functionality. Every time of processing a new json, if already existed, will delete all the test list under a Test schedule. so everytime of sample creation, the latest test schedule (the grouping of Analysis) will be assigned.

Commented [NR237]: As of now only QUAN-01, but in future it can change for Dragon plant as other plants.

Commented [XW238R237]: Noted.

Commented [NR239]: QUAN-05

Commented [XW240R239]: Updated.

Commented [NR241]: It should be available in QCLAD in Phase B Go Live

Commented [XW242R241]: Please refer to another alignment email that I sent for this matter.

If that is aligned, then TF is able to include this requirement without change.

If not as what is understood, then we need to discuss further.

Commented [NR243]: Please ensure skip lot testing is considered

Commented [XW244R243]: The skip lot requirement will be considered when processing the inspection lot.

Please pay attention to this part and confirm the understanding is correct:

5. **As per the skipping testing, each inspection lot w inspection or not. Hence, QCLAD needs to furthe**

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- if it is equal to "1", which means current MIC shou the Test/result, and then on the result table, set tl
- If equal to "4", further fetch the "pmethode" to lo of this component (MIC) to "Entered", and set set

Commented [NR245R243]: Yes it is fine

Commented [NR246]: New MLP to be used for new inspection lot and older inspection lots need to use the older inspection plan and thereby older specification limits.

Commented [XW247R246]: Yes. This is confirmed

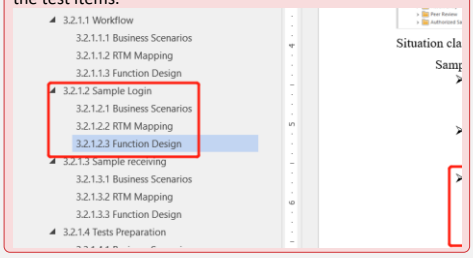
11. **At the time of user manually registering sample for case such JIT/Truck, user has to provide the "plant code" +"IDH code" + "Activity No" so that QCLAD can auto find the unique Test schedule and MLP to assign to the sample.**

Commented [NR248]: Is this applicable for non standard testing which has an inspection plan in SAP.

Commented [XW249R248]: Not included for the time being.

For non-standard testing, the requirement to auto map with test schedule and MLP when processing the excel file, is already logged as a UAT issue.

It is considered as a Change.
In prior FS chapter in detailing non-standard testing workflow, it is agreed that user needs to manually decide the test items.



~~3.2.5.1.1~~3.2.6.1 *SAP to QCLAD Inspection Lot Transaction Data*

~~3.2.5.1.1~~3.2.6.1.1 *Introduction*

Henkel's testing workflows, if divided by the triggering point, can be categorized into two group. One is that SAP triggers the inspection lot down and register the sample in QCLAD; the other one is that QCLAD firstly registers the sample either manually by user, such for the case of JIT/Truck workflow; or automatically by processing the excel file, such for the non-standard testing workflow.

When SAP push down the inspection lot, it is required that QCLAD registers the sample and auto assign the test schedules (the set of Test method need to inspect) and MLP (the spec of all the testing parameters) to it based on the given information such as material code, plant code and activity no.

~~3.2.5.1.2~~3.2.6.1.2 *Interface URL address*

https://{QCLAD domain name}:56105/SAPtoLIMS/TransactionData/InspectionLot

Notes: the full URL address is up for changes, above is for reference.

~~3.2.5.1.3~~3.2.6.1.3 *Field Mapping*

Please refer to json file sent along with confirmation email.

~~Refers to~~

~~3.2.5.1.5~~3.2.6.1.4 *Exampled transaction json*

Please refer to json file sent along with confirmation email.

~~Refers to~~

~~3.2.5.1.7~~3.2.6.1.5 *Exampled response json*

Please refer to chapter 3.2.6.4.5 Exampled response json

~~3.2.5.1.8~~3.2.6.1.6 *Frequency*

As frequency defined in the scheduled job in SAP.

~~3.2.5.1.9~~3.2.6.1.7 *System Logic*

1. **For each Inspection lot, it could contain more than one Activity No (Phase no), that is corresponding to the number of sample, for example IPQC and OQC sample , to be inspected. Hence, QCLAD should either newly create or update a sample based on the combination of "plant code" +"Material code"+"batch no"+ "activity no". if matched, then update the field information into the sample; if not, then create one.**

2. **Each inspection lot json would provide a status.**

- **If A, QCLAD would create sample accordingly.**
- **If D or C, QCLAD would cancel the sample/Test/Result already created completely.**

3. **When creating a sample per each activity no, QCLAD will save the information "Inspection lot no", "UUID", "IDH code" +"Plant code" +"Activity No" +"Batch No" +"shelf life expiration" +"next inspection date"+"Manufacturing date" provided by interface.**

Commented [NR250]: For several cases of bulk, FG testing there wont be manufacturing date, expiration date. System has to populate as it comes.

Commented [XW251R250]: It is covered in the Batch Information interface.

For inspection lot interface, when processing, just store as it is. If empty, then empty in QCLAD in the first place.

4. **Based on the given combination of "plant code" +"material code"+ "Activity no" , QCLAD will auto query and match the unique MLP and Test Schedule.**

- Once a Test Schedule is assigned to the sample, each Analysis within that Test Schedule will be created as a Test when assigning to sample. And if any component list is specified of an Analysis, only the selected component(MIC) of that component list of that Analysis will be created as Result when assigned to the sample. If there is none component list specified, all the components of that Analysis will be created Result when assigned to the sample.
- The value of "result type", "Unit" and "decimal place" will be always fetched from the component table of each Analysis into each Result respectively when assigning to sample.
- For the COA type of Analysis, since there is none component list defined in the Test Schedule, the "SAP order no", "SAP MIC code", "short text" and "info field 3" will be fetched directly from component table of each Analysis into each Result respectively.
- For the non COA type of Analysis, since there is component list defined in the Test Schedule, the "order no", "SAP MIC code", "short text" and "info field 3" will be fetched from component list table of each Analysis into each Result respectively.

5. **As per the skipping testing, each inspection lot would provide the indicator on the MIC level if requires inspection or not. Hence, QCLAD needs to further evaluate each result (MIC).**

- if it is equal to "1", which means current MIC should be inspected, then follow standard process to create the Test/result, and then on the result table, set the value to "Hen_Required (Boolean field)"as TRUE.
- If equal to "4", further fetch the "pmethode" to locate the actual Analysis and then to update the status of this component (MIC) to "Entered", and set set the value to "Hen_Required (Boolean field)"as FALSE.

~~3.2.5.2~~3.2.6.2 QCLAD to SAP Inspection result transaction data

~~3.2.5.2.1~~3.2.6.2.1 Introduction

For any workflow of which SAP has created inspection lot, their raw results, once authorized in QCLAD, should be interfaced back to SAP. And then SAP will perform the usage decision and generate COA accordingly. Otherwise, for case such as non-standard testing, the results can remain inside QCLAD.

For some cases such as Finished Product Inspection (OQC), the sample is registered by inspection lot from SAP first and links with a SAP inspection lot No in the very beginning, then at the time of this subjected sample's authorization, its results should be automatically sent to SAP.

Otherwise for cases such as JIT/Truck workflow where sample is firstly registered in QCLAD and without a SAP inspection lot No immediately, at the time of such sample's authorization, if still without inspection lot no, would be not auto pushing its' result to SAP. Once inspection lot is created for that JIT/Truck sample and links, its result then will get sent to SAP automatically.

~~3.2.5.2.2~~3.2.6.2.2 Interface URL address

https://{QCLAD domain name}:56105/LIMStoSAP/TransactionData/InspectionResult

Notes: the full URL address is up for changes, above is for reference.

~~3.2.5.2.3~~3.2.6.2.3 Field Mapping

Please refer to json file sent along with confirmation email.

~~Refers to~~

~~3.2.5.2.5~~3.2.6.2.4 Exampled transaction json

Please refer to json file sent along with confirmation email.

~~Refers to~~

~~3.2.5.2.7~~3.2.6.2.5 Exampled response json

Please refer to chapter 3.2.6.4.5 Exampled response json

~~3.2.5.2.8~~3.2.6.2.6 Frequency

As frequency defined in the scheduled background task in QCLAD.

The frequency is configurable.

~~3.2.5.2.9~~3.2.6.2.7 System Logic

1. QCLAD would add two fields on sample tables, and the logic to populate the value as below:
   to_be_sent: when current sample has a "SAP inspection lot no" and its status is "authorized", then the value of this field is set as TRUE, meaning the result of this sample is ready to be sent out to SAP on next background task occurrence.
   Has_sent: when current sample's results has successfully sent to SAP, then the value of this field is set as TRUE.

2. For each transaction, under each sample, only the results which has a SAP MIC code will be included in the json.
3. Test status：A； Result：A and has MIC Code

Commented [NR252]: URL will be provided by Henkel

Commented [NR253]: Kindly modify the JSON file as request sent by Ashu on 12.11.2024

Commented [NR254]: Please provide the frequency

Commented [XW255R254]: The frequency is configurable.

Commented [AS256]: Scenarios for result update in SAP:

1- Inspection lot is created and sent to QCLAD ,sample result will be pushed back to SAP .
1- Inspection lot is created and sent to QCLAD and later it got cancelled in that case SAP wont require resluts back for the cancelled Inspection Lot .

Commented [XW257R256]: Sure. The inspection is cancelled, the sample/test/result prior created for that inspection lot will be all cancelled completely.